

Designing the TechNova Solutions Campus Network

An End-to-End Case Study in Computer Networks & Data Communication

Combined Design, Configuration, Verification & Evidence Report

Document structure

This version combines the written case study, practical Wireshark/macOS verification, Cisco Packet Tracer configuration records, routing/DHCP/NAT results, STP/RSTP redundancy testing, and final network verification into the original case-study unit structure. The material is organized by networking topic rather than by phase.

0. The Scenario: Company Profile & Requirements

TechNova Solutions Pvt. Ltd. is moving into its own three-floor office building and needs a reliable network for around 150 staff and guests. The network should keep departments separated, provide shared internet access, handle failures smoothly, and support future growth. It must also support VoIP phones and wireless devices.

Floor	Departments	Approx. Users	Special Needs
Ground Floor	Reception, Sales & Marketing, Guest Lounge	45	Guest Wi-Fi; high visitor footfall
First Floor	Engineering / Development, QA	65	Heavy LAN traffic; code repositories; build servers
Second Floor	HR, Finance & Accounts, Management, Server/IT Room	40	Sensitive data; restricted access; core servers

0.1 Business Requirements → Design Consequences

Business requirement	Design consequence
Reliable, fast local connectivity	Managed Ethernet switches, Cat6 endpoint cabling and high-capacity fibre backbone links.
Logical separation between departments	Separate VLANs and IP subnets for departments, servers, voice, guest and management traffic.
Centralized internet access	Layer-3 core plus edge router/firewall for ISP connectivity, NAT and security.

Resilience	Redundant uplinks, RSTP and EtherChannel; remaining single points of failure identified.
30–40% room for growth	CIDR subnet sizing, spare switch ports, scalable backbone and address space.
VoIP and wireless support	Dedicated voice VLAN and wireless APs with separate guest service.

0.2 Requirement Traceability

Requirement	Where it is addressed
Reliable connectivity	Topology, Devices & Media, IP Addressing, Core Protocols, Verification, Switching
Departmental isolation	IP Addressing, VLANs, Trunking and Inter-VLAN Routing
Centralized internet access	Topology, edge routing, NAT and final network build
Resilience	Redundant topology, STP/RSTP, EtherChannel and failover verification
30–40% growth	Subnetting, fibre backbone, switch capacity and scalable VLAN structure
VoIP / Wireless	Devices & Media, voice VLAN and guest wireless design

0.2Mini Case Study — Arden Technologies

Arden Technologies is a growing software and digital-services company with a three-floor office and around 120 employees. Its teams include Engineering, Sales, HR, Finance, Management and IT. The company needs reliable wired and wireless access for computers, laptops, VoIP phones, printers and internal services.

Engineering needs high-speed access to repositories and build servers, while sensitive departments must be separated from general and guest traffic. A central router/firewall, VLANs, Cat6 cabling and a fibre backbone are used to support these requirements.

1. Network Fundamentals, Types & Topology

1.1 Network and Topology Choice

For TechNova, a hybrid or extended-star topology makes the most sense. Each floor works around its access switch like a normal star network, while the core has redundant links for better reliability. In simple terms, traffic follows the path Internet/ISP ? Edge Router ? Layer-3 Core Switch ? Floor Access Switches ? PCs, APs, phones and other devices.

Topology	Why it is a worse fit for TechNova
Bus	A backbone/cable failure can disrupt many users; collisions and troubleshooting become problematic as the network grows.
Ring	A broken link or device can disrupt the ring unless extra redundancy is introduced.
Full Mesh	High redundancy, but excessive cabling, switch ports, cost and management overhead at this scale.
Hybrid / Extended-Star	Chosen because star access wiring is manageable while redundant core links provide fault tolerance.

1.2 Remaining Single Point of Failure

One important weakness is that the Layer-3 core switch is still a single point of failure. In a real production network, this could be improved by adding a second core switch with high-availability features and dual uplinks.

Implemented topology evidence

The Packet Tracer work established redundant switch connectivity, an edge router, access switching, server/IT connectivity and a triangular Layer-2 path used by RSTP for loop protection.

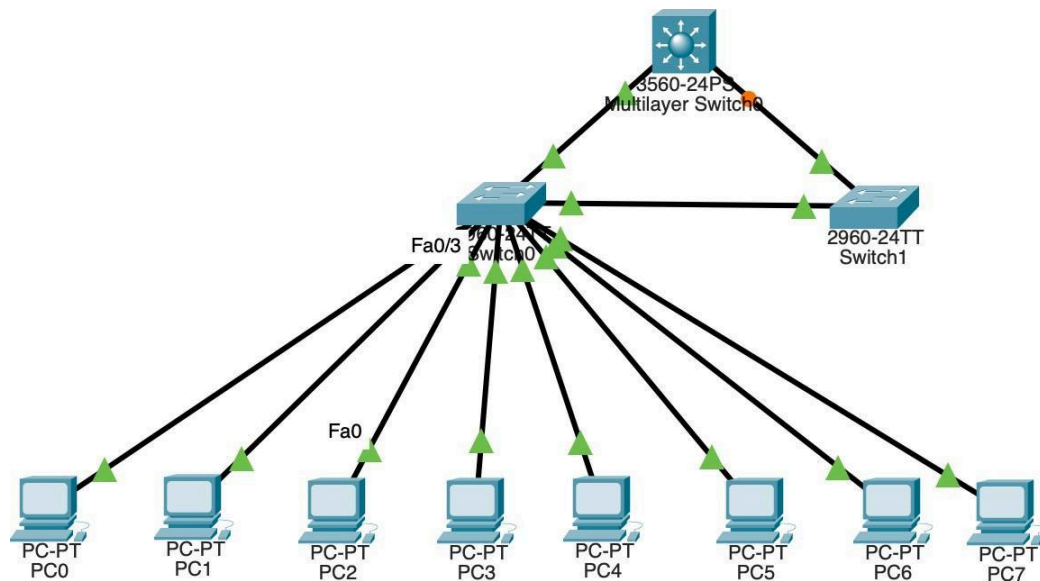


Figure 1 — Verified redundant-switch topology used for RSTP loop protection.

2. The OSI Model, TCP/IP Model & Encapsulation

The layered approach is useful both when designing the network and when finding problems. A switch mainly works at Layer 2 and uses MAC addresses to move Ethernet frames, while a router or Layer-3 switch works at Layer 3 and uses IP addresses to move packets between networks.

2.1 OSI vs TCP/IP

OSI 7-layer	TCP/IP 5-layer	Classic TCP/IP 4-layer
Application	Application	Application
Presentation	Application	Application
Session	Application	Application
Transport	Transport	Transport
Network	Network	Internet
Data Link	Data Link	Network Access
Physical	Physical	Network Access

2.2 Practical Wireshark Capture

For the practical part, Wireshark 4.6.7 was run on the macOS Wi-Fi interface en0. Network traffic was generated and captured, and Frame 36 was selected for closer inspection. The frame showed the path Ethernet ? IPv4 ? TCP ? TLS, including a TLS 1.3 Client Hello.

Capture item	Actual result
Wireshark version	4.6.7
Interface	Wi-Fi — en0
Selected frame	Frame 36
Protocol stack	Ethernet → IPv4 → TCP → TLS
Application protocol	TLS 1.3
TLS message	Handshake Protocol: Client Hello
SNI	mobile.events.data.microsoft.com
Packets captured	2924
Packets displayed by tcp filter	1392
Dropped packets	0

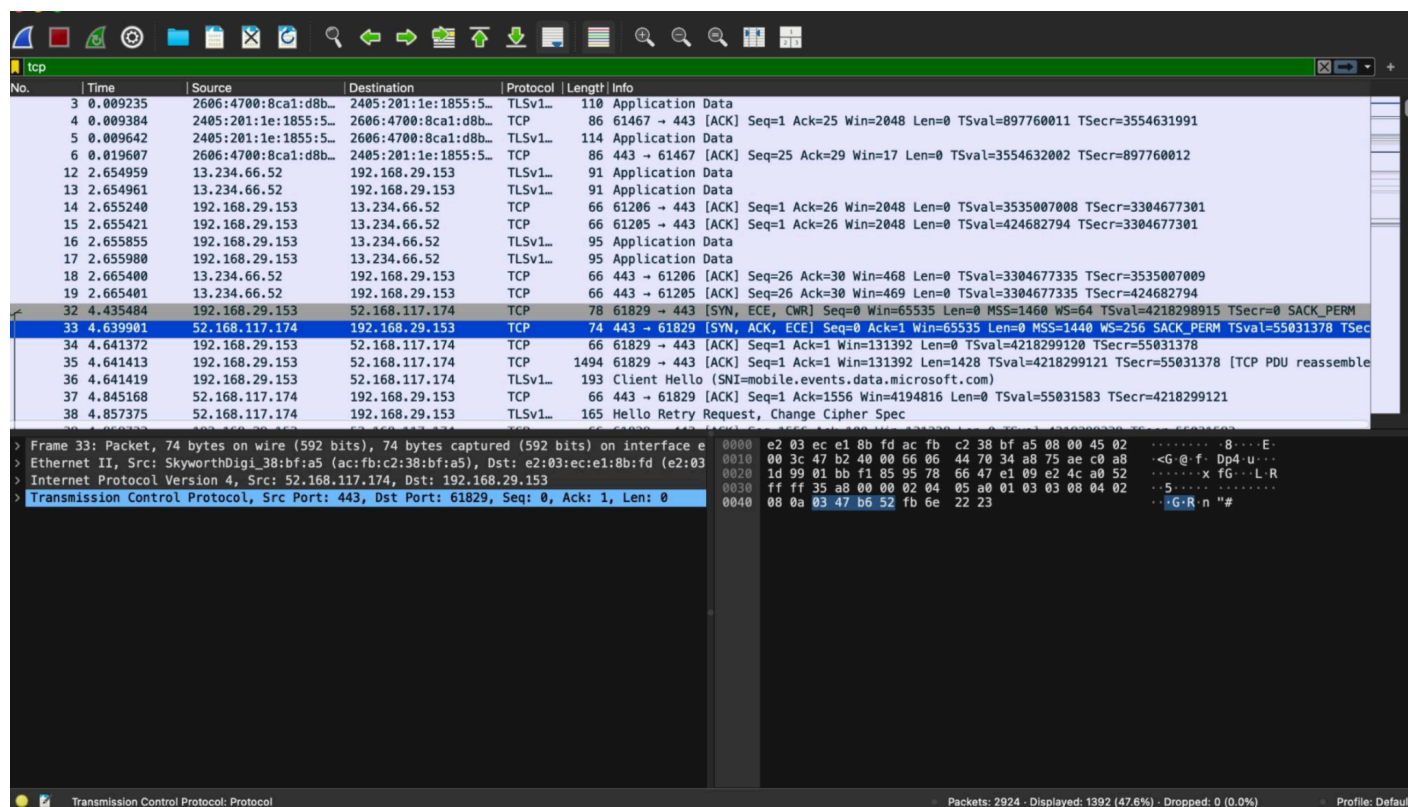


Figure 2 — Wireshark start screen showing the en0 capture interface.

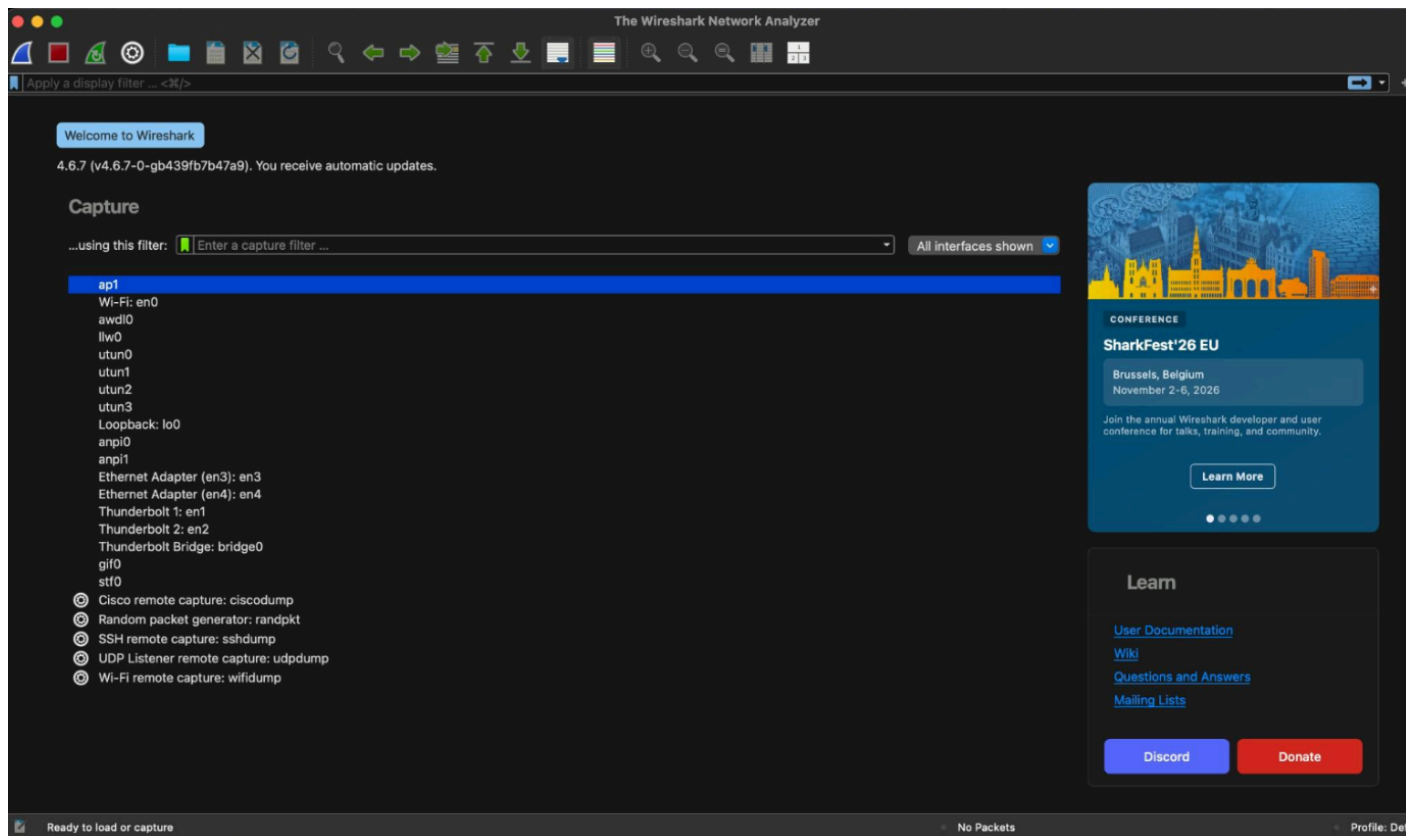


Figure 3 — Wireshark TCP capture on the Wi-Fi interface.

2.3 Frame 36 — Encapsulation Analysis

Layer	Field	Actual value
Frame	Frame number	36
Frame	Frame length	193 bytes (1544 bits)
Frame	Interface	en0
Data Link	Source MAC	e2:03:ec:e1:8b:fd
Data Link	Destination MAC	ac:fb:c2:38:bf:a5
Data Link	EtherType	IPv4 (0x0800)
Network	Source IP	192.168.29.153
Network	Destination IP	52.168.117.174
Network	IP total length	179 bytes
Network	Protocol	TCP (6)
Network	TTL	64
Transport	Source port	61829
Transport	Destination port	443
Transport	TCP payload	127 bytes
Application	Protocol	TLS 1.3
Application	Message	Handshake Protocol: Client Hello

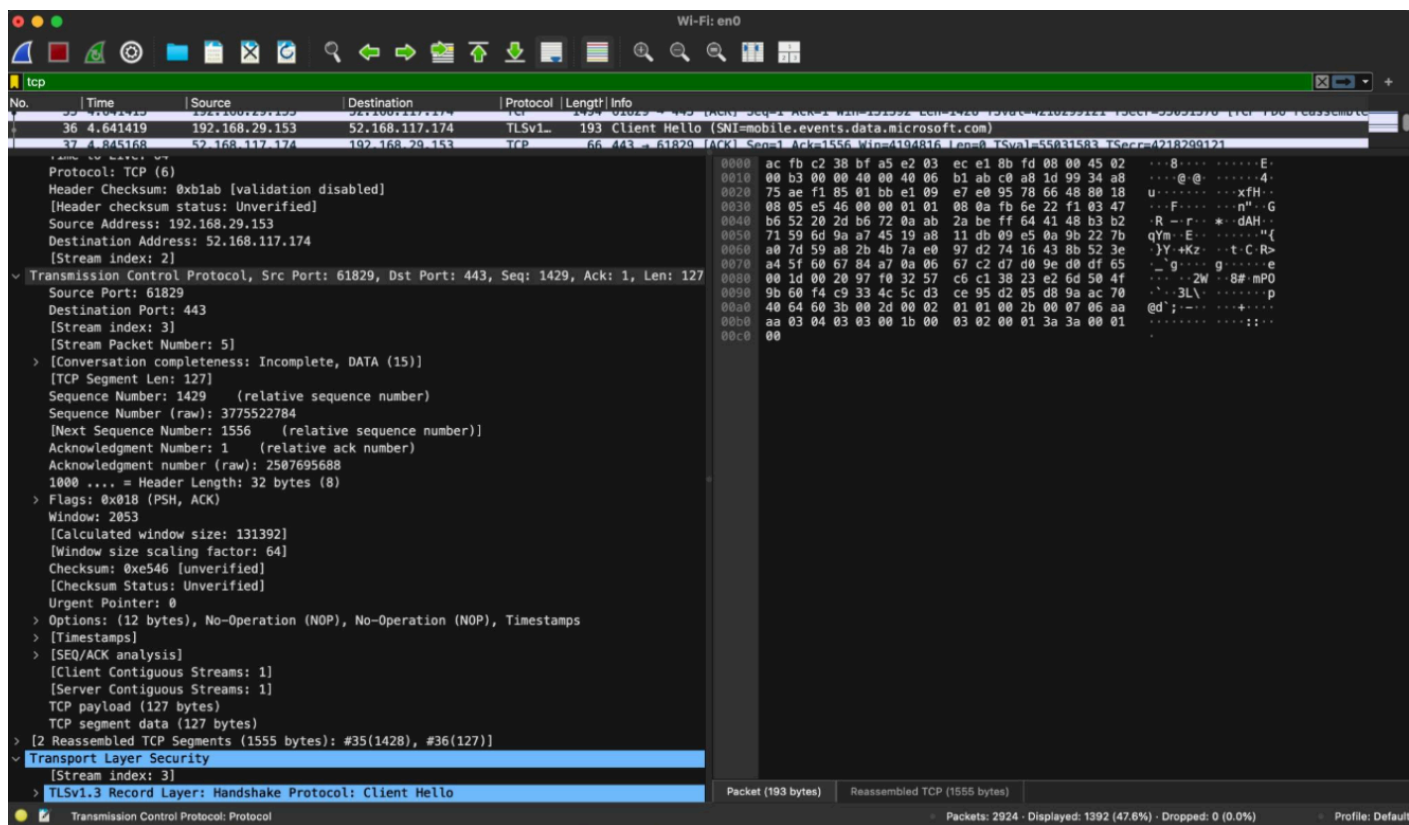


Figure 4 — Frame 36 Ethernet and IPv4 details.

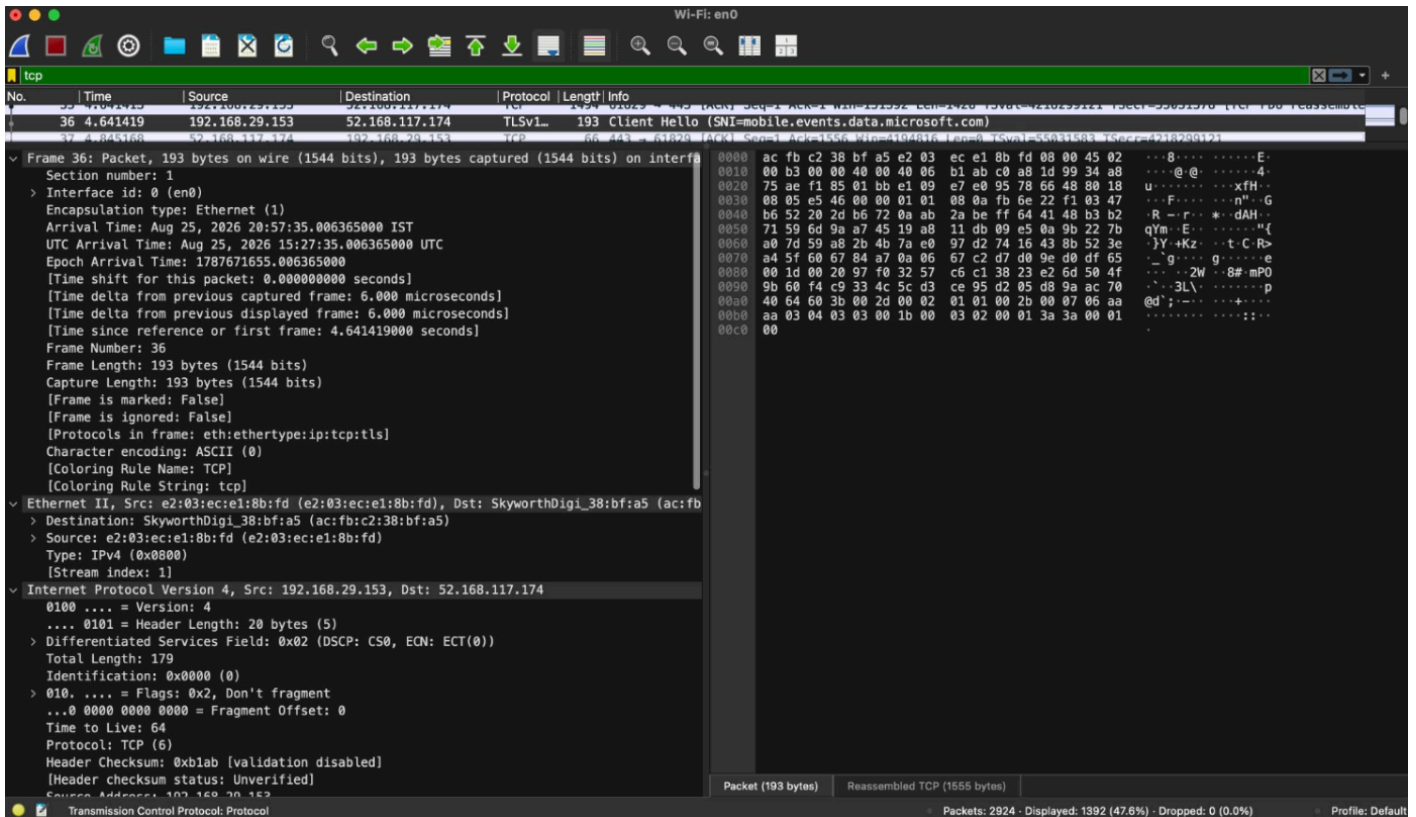


Figure 5 — Frame 36 TCP and TLS 1.3 Client Hello details.

2.4 Encapsulation Chain

PDU	Layer	Captured contents
Data	Application	TLS 1.3 Client Hello; SNI mobile.events.data.microsoft.com
Segment	Transport	TCP; source 61829 → destination 443; payload 127 bytes
Packet	Network	IPv4; 192.168.29.153 → 52.168.117.174; total length 179 bytes
Frame	Data Link	Ethernet; e2:03:ec:e1:8b:fd → ac:fb:c2:38:bf:a5
Bits	Physical	193-byte Ethernet frame = 1544 bits

3. Networking Devices & Transmission Media

Device	Primary OSI layer	Role / justification
Repeater / media converter	Layer 1	Regenerates or converts physical signals; useful for longer backbone media.
Hub	Layer 1	Repeats bits to all ports; not used in the production design.
Bridge	Layer 2	Uses MAC addresses to forward frames.
Switch	Layer 2	Learns MAC addresses and forwards Ethernet frames.
Layer-3 switch	Layer 3	Provides routing between VLANs through SVIs.
Router	Layer 3	Routes packets between IP networks and provides the edge path.
Access Point	Layers 1/2	Provides wireless access and bridges wireless clients to the LAN.
Firewall / Gateway	Multiple	Filters traffic and can provide NAT and security functions.

3.1 Transmission Media

Medium	Typical distance	Typical bandwidth	TechNova use
Cat6 UTP	Up to 100 m channel	1 Gbps at 100 m; 10 Gbps for shorter runs	Office endpoints
Cat6A STP	Up to 100 m channel	10 Gbps	High-speed / EMI-sensitive copper links
Multimode fibre	Hundreds of metres depending on OM grade	1–100+ Gbps depending on grade/transceiver	Inter-floor backbone

Fibre is selected for the inter-floor backbone because it provides high bandwidth, longer reach and immunity to electromagnetic interference.

3.2T568B Cable Termination

Pin	T568B conductor
1	White/Orange
2	Orange
3	White/Green
4	Blue
5	White/Blue
6	Green
7	White/Brown
8	Brown

Practical status — physical cabling

The Cat6 termination and cable-tester tasks were not physically performed because the required cable kit, RJ-45 connectors/crimping tool and tester were unavailable. No fabricated photograph or tester result is claimed.

4. IP Addressing & Subnetting Plan

The case-study design uses the private 10.10.0.0/16 block and CIDR subnetting. The subnet sizes are intentionally different because departments have different user populations.

VLAN	Department	Design subnet	Mask	Usable hosts	Host range
10	HR / Admin	10.10.10.0/27	255.255.255.224	30	10.10.10.1–10.10.10.30
20	Finance & Accounts	10.10.20.0/27	255.255.255.224	30	10.10.20.1–10.10.20.30
30	Sales & Marketing	10.10.30.0/26	255.255.255.192	62	10.10.30.1–10.10.30.62
40	Engineering / QA	10.10.40.0/25	255.255.255.128	126	10.10.40.1–10.10.40.126
50	Servers / IT	10.10.50.0/28	255.255.255.240	14	10.10.50.1–10.10.50.14
60	VoIP phones	10.10.60.0/25	255.255.255.128	126	10.10.60.1–10.10.60.126
70	Guest Wi-Fi	10.10.70.0/24	255.255.255.0	254	10.10.70.1–10.10.70.254
99	Management	10.10.99.0/28	255.255.255.240	14	10.10.99.1–10.10.99.14

4.1 Sales & Marketing Subnet Calculation

```

Given 10.10.30.0/26:
Host bits = 32 - 26 = 6
Total addresses = 2^6 =
64 Usable hosts = 64 - 2
= 62 Network address =
10.10.30.0
Broadcast address = 10.10.30.63
Valid host range = 10.10.30.1–10.10.30.62
Subnet mask = 255.255.255.192

```

4.2 Practical IP/CIDR Self-Quiz

#	IP/CIDR	Subnet mask	Network address	Usable hosts
1	10.10.10.5/27	255.255.255.224	10.10.10.0	30
2	10.10.20.14/27	255.255.255.224	10.10.20.0	30
3	10.10.30.25/26	255.255.255.192	10.10.30.0	62
4	10.10.40.55/25	255.255.255.128	10.10.40.0	126
5	10.10.50.10/28	255.255.255.240	10.10.50.0	14

6	10.10.60.80/25	255.255.255.128	10.10.60.0	126
7	10.10.70.200/24	255.255.255.0	10.10.70.0	254
8	10.10.99.8/28	255.255.255.240	10.10.99.0	14
9	10.10.254.1/30	255.255.255.252	10.10.254.0	2
10	192.168.1.70/26	255.255.255.192	192.168.1.64	62

Addressing distinction

The verified Packet Tracer build used a separate laboratory addressing plan: VLANs 10/20/30/40 were implemented as 192.168.10.0/24, 192.168.20.0/24, 192.168.30.0/24 and 192.168.40.0/24. This is retained as an implementation record and is not presented as a replacement for the original 10.10.0.0/16 design plan.

5. Core Protocols: Ethernet, ARP, ICMP, DHCP & DNS

5.1 ARP — Actual Capture

The Wireshark ARP capture used Frames 153 and 154. The observed exchange involved the local gateway 192.168.29.1 and the Mac at 192.168.29.153.

Field	Frame 153 — Request	Frame 154 — Reply
Ethernet source	ac:fb:c2:38:bf:a5	e2:03:ec:e1:8b:fd
Ethernet destination	ff:ff:ff:ff:ff:ff (broadcast)	ac:fb:c2:38:bf:a5
Sender IP	192.168.29.1	192.168.29.153
Target IP	192.168.29.153	192.168.29.1
Operation	Request	Reply
Message	Who has 192.168.29.153? Tell 192.168.29.1	192.168.29.153 is at e2:03:ec:e1:8b:fd

Wireshark packet capture showing ARP traffic. The packet list at the top shows several ARP requests. Frame 153 is highlighted, showing an ARP request from SkyworthDigi_38:bf:a5 to e2:03:ec:e1:8b:fd. The packet details pane shows the following information:

- Section number: 1
- Interface id: 0 (en0)
- Encapsulation type: Ethernet (1)
- Arrival Time: Aug 25, 2026 21:10:53.340684000 IST
- UTC Arrival Time: Aug 25, 2026 15:40:53.340684000 UTC
- Epoch Arrival Time: 1787672453.340684000
- [Time shift for this packet: 0.000000000 seconds]
- [Time delta from previous captured frame: 36.000 microseconds]
- [Time delta from previous displayed frame: 36.000 microseconds]
- [Time since reference or first frame: 9.286504000 seconds]
- Frame Number: 154
- Frame Length: 42 bytes (336 bits)
- Capture Length: 42 bytes (336 bits)
- [Frame is marked: False]
- [Frame is ignored: False]
- [Protocols in frame: eth:ethertype:arp]
- Character encoding: ASCII (0)
- [Coloring Rule Name: ARP]
- [Coloring Rule String: arp]
- Ethernet II, Src: e2:03:ec:e1:8b:fd (e2:03:ec:e1:8b:fd), Dst: SkyworthDigi_38:bf:a5 (ac:fb:c2:38:bf:a5)
 - Destination: SkyworthDigi_38:bf:a5 (ac:fb:c2:38:bf:a5)
 - Source: e2:03:ec:e1:8b:fd (e2:03:ec:e1:8b:fd)
 - Type: ARP (0x0006)
 - [Stream Index: 2]
- Address Resolution Protocol (reply)

The packet bytes pane shows the following hexadecimal data:

```

0000  ac fb c2 38 bf a5 e2 03  ec e1 8b fd 08 06 00 01  ...8...
0010  08 00 06 04 00 02 e2 03  ec e1 8b fd c0 a8 1d 99  ...8...
0020  ac fb c2 38 bf a5 c0 a8  1d 01

```

Figure 6 — ARP Request, Frame 153.

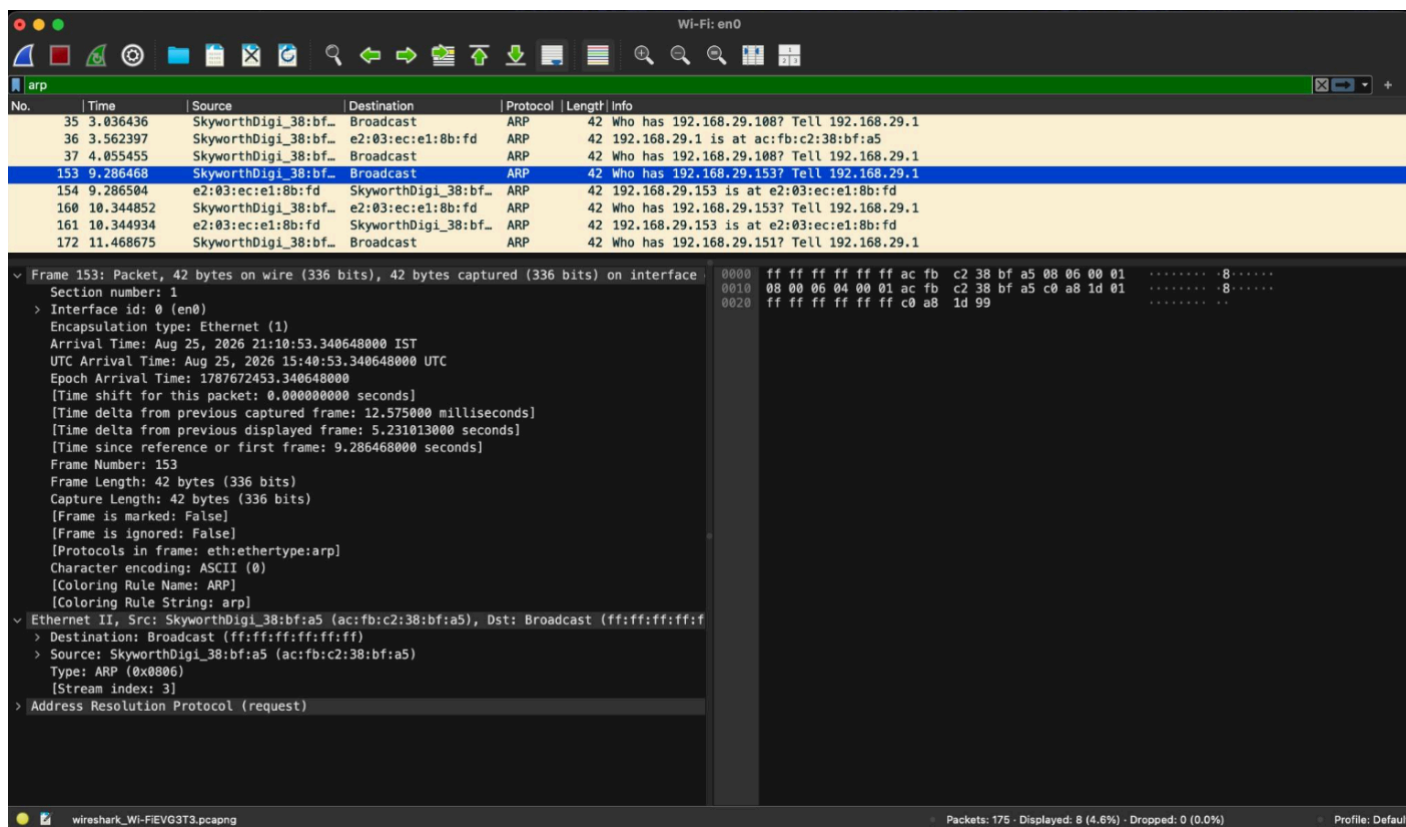


Figure 7 — ARP Reply, Frame 154.

The ARP request is sent as a broadcast because the sender does not yet know the MAC address of the device it wants to reach. Once the target responds, the reply can be sent directly back to the requesting device.

5.2DHCP — Verified Packet Tracer Service

In the Packet Tracer setup, the multilayer switch acts as the DHCP server for VLANs 10, 20, 30 and 40. Each DHCP pool matches its VLAN network and uses the SVI as the gateway, with 8.8.8.8 configured as DNS. The first few addresses were excluded so they could be kept for gateways and other fixed devices.

VLAN	Network	Gateway	DNS	Observed clients
10	192.168.10.0/24	192.168.10.1	8.8.8.8	PC0 .10, PC1 .11
20	192.168.20.0/24	192.168.20.1	8.8.8.8	PC2 .10, PC3 .11
30	192.168.30.0/24	192.168.30.1	8.8.8.8	PC4 .10, PC5 .11
40	192.168.40.0/24	192.168.40.1	8.8.8.8	PC6 .10, PC7 .11



Figure 8 — DHCP pool configuration and verification.

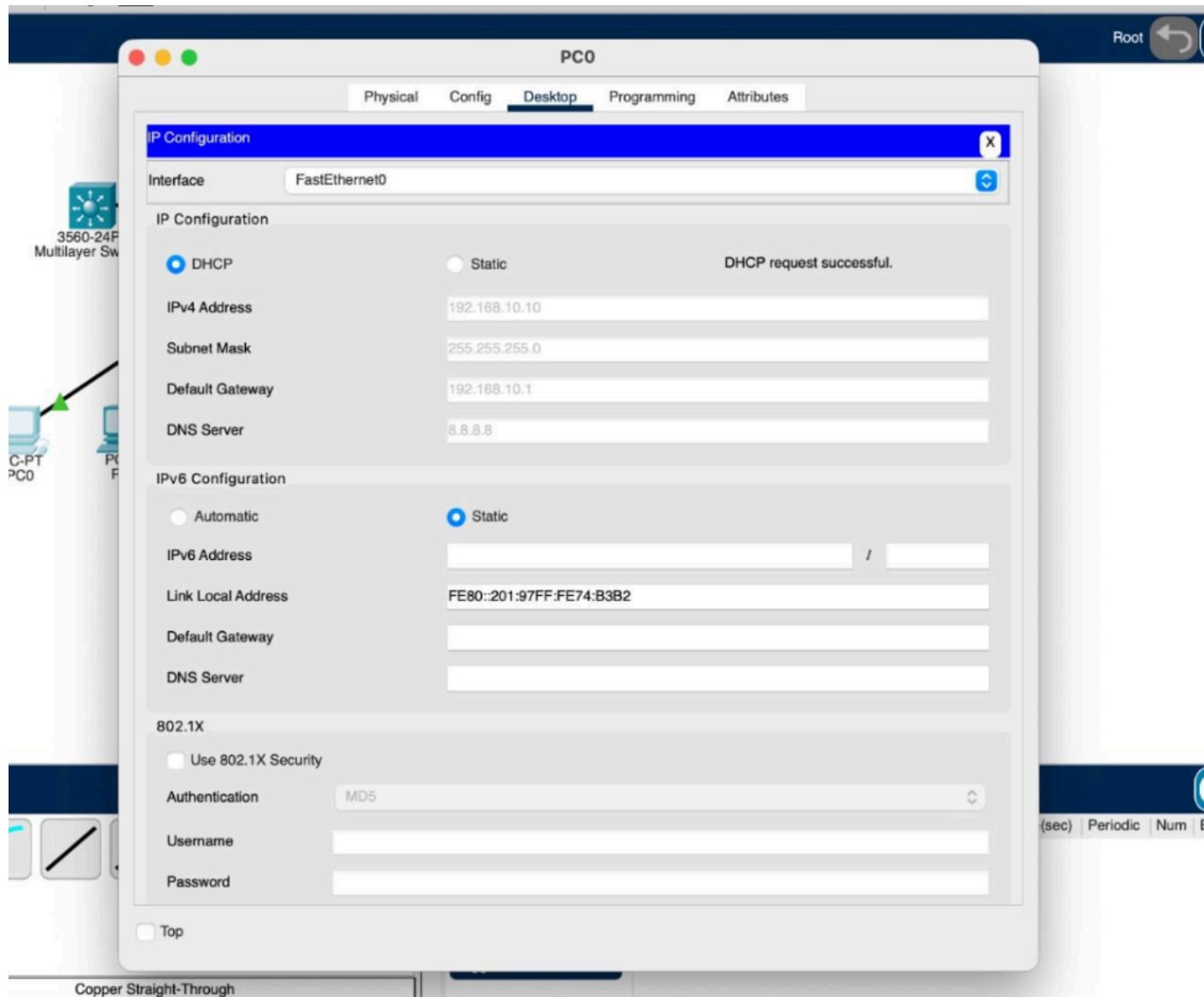


Figure 9 — PC DHCP request/address acquisition.



Figure 10 — DHCP binding table showing eight automatic leases.

5.3DNS

The written design includes a local DNS record such as hrms.technova.local pointing to 10.10.50.10 and suggests checking it with nslookup or dig. However, the practical evidence provided does not contain an actual DNS A-record capture, so no extra DNS result is claimed.

5.4ICMP

ICMP Echo Request/Reply was used throughout the practical verification. Successful Packet Tracer tests returned four replies with 0% loss; the macOS tests below provide measured LAN and internet RTT.

6. Verification & Practical Exercises

6.1macOS IP Configuration

Parameter	Actual result
Interface	Wi-Fi (en0)
Configuration	DHCP
IPv4 address	192.168.29.153

Subnet mask	255.255.255.0
CIDR	/24
Default gateway/router	192.168.29.1
DHCP server	192.168.29.1
DNS server	192.168.29.1
Broadcast	192.168.29.255
MAC address	e2:03:ec:e1:8b:fd

```

> ipconfig getifaddr en0
192.168.247.148
> networksetup -getinfo Wi-Fi
DHCP Configuration
IP address: 192.168.247.148
Subnet mask: 255.255.255.0
Router: 192.168.247.20
Client ID:
IPv6: Automatic
IPv6 IP address: none
IPv6 Router: none
Wi-Fi ID: 2c:ca:16:7f:51:af
> scutil --dns | grep 'nameserver\[0-9\]*\]'
nameserver[0] : 2401:4900:abad:c954::ae
nameserver[1] : 192.168.247.20
nameserver[0] : 2401:4900:abad:c954::ae
nameserver[1] : 192.168.247.20

> ipconfig getpacket en0
op = BOOTREPLY
htype = 1
flags = 0x0
hlen = 6
hops = 0
xid = 0xc880586d
secs = 0
ciaddr = 0.0.0.0
yiaddr = 192.168.247.148
siaddr = 192.168.247.20
giaddr = 0.0.0.0
chaddr = 12:48:22:c7:a7:6a
sname =
file =
options:
Options count is 11
dhcp_message_type (uint8): ACK 0x5
server_identifier (ip): 192.168.247.20
lease_time (uint32): 0xe0f
renewal_time_value (uint32): 0x707
rebinding_time_value (uint32): 0xcdd
subnet_mask (ip): 255.255.255.0
broadcast_address (ip): 192.168.247.255
router (ip_mult): {192.168.247.20}
domain_name_server (ip_mult): {192.168.247.20}
vendor_specific (opaque):
0000 41 4a 04 12 4f 49 44 5f 4d 45 54 45 52 45 44  ANDROID_METERED

end (none):

```

Figure 11 — macOS IP configuration, subnet mask, router and DNS information.

```

> ipconfig getifaddr en0
192.168.247.148
> networksetup -getinfo Wi-Fi
DHCP Configuration
IP address: 192.168.247.148
Subnet mask: 255.255.255.0
Router: 192.168.247.20
Client ID:
IPv6: Automatic
IPv6 IP address: none
IPv6 Router: none
Wi-Fi ID: 2c:ca:16:7f:51:af
> scutil --dns | grep 'nameserver\[0-9\]*\]'
nameserver[0] : 2401:4900:abad:c954::ae
nameserver[1] : 192.168.247.20
nameserver[0] : 2401:4900:abad:c954::ae
nameserver[1] : 192.168.247.20

```

Figure 12 — DHCP packet information from the macOS environment.

The observed 192.168.29.0/24 network is the student's actual local Wi-Fi network and is separate from the TechNova Packet Tracer addressing.

6.2 Ping Verification

Metric	Local gateway 192.168.29.1	Internet host 8.8.8.8
Packets transmitted	10	10
Packets received	10	10
Packet loss	0.0%	0.0%
Minimum RTT	4.431 ms	10.813 ms
Average RTT	7.398 ms	15.324 ms
Maximum RTT	11.431 ms	23.046 ms
Standard deviation	2.716 ms	4.236 ms


```

> ping -c 10 192.168.29.1
PING 192.168.29.1 (192.168.29.1): 56 data bytes
64 bytes from 192.168.29.1: icmp_seq=0 ttl=64 time=10.054 ms
64 bytes from 192.168.29.1: icmp_seq=1 ttl=64 time=5.796 ms
64 bytes from 192.168.29.1: icmp_seq=2 ttl=64 time=4.431 ms
64 bytes from 192.168.29.1: icmp_seq=3 ttl=64 time=11.409 ms
64 bytes from 192.168.29.1: icmp_seq=4 ttl=64 time=11.431 ms
64 bytes from 192.168.29.1: icmp_seq=5 ttl=64 time=8.363 ms
64 bytes from 192.168.29.1: icmp_seq=6 ttl=64 time=4.872 ms
64 bytes from 192.168.29.1: icmp_seq=7 ttl=64 time=4.732 ms
64 bytes from 192.168.29.1: icmp_seq=8 ttl=64 time=4.607 ms
64 bytes from 192.168.29.1: icmp_seq=9 ttl=64 time=8.286 ms

--- 192.168.29.1 ping statistics ---
10 packets transmitted, 10 packets received, 0.0% packet loss
round-trip min/avg/max/stddev = 4.431/7.398/11.431/2.716 ms

> ping -c 10 8.8.8.8
PING 8.8.8.8 (8.8.8.8): 56 data bytes
64 bytes from 8.8.8.8: icmp_seq=0 ttl=112 time=14.203 ms
64 bytes from 8.8.8.8: icmp_seq=1 ttl=112 time=10.813 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=112 time=16.040 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=112 time=11.904 ms
64 bytes from 8.8.8.8: icmp_seq=4 ttl=112 time=15.908 ms
64 bytes from 8.8.8.8: icmp_seq=5 ttl=112 time=14.858 ms
64 bytes from 8.8.8.8: icmp_seq=6 ttl=112 time=13.532 ms
64 bytes from 8.8.8.8: icmp_seq=7 ttl=112 time=18.033 ms
64 bytes from 8.8.8.8: icmp_seq=8 ttl=112 time=23.046 ms
64 bytes from 8.8.8.8: icmp_seq=9 ttl=112 time=14.903 ms

--- 8.8.8.8 ping statistics ---
10 packets transmitted, 10 packets received, 0.0% packet loss
round-trip min/avg/max/stddev = 10.813/15.324/23.046/3.236 ms

> █

```

Figure 13 — Local gateway and remote 8.8.8.8 ping results.

The local gateway showed a lower average response time of 7.398 ms, while 8.8.8.8 averaged 15.324 ms. Both tests completed successfully with 0% packet loss.

6.3 Traceroute

Hop	Observed address(es)	Observation
1	192.168.29.1	Local default gateway
2	10.38.152.1	Private upstream network
3	172.31.0.238; 172.31.2.26	Private upstream routing
4	172.28.75.32; 192.168.53.188; 172.28.75.26	Private upstream routing
5	172.26.76.214	Private upstream routing
6	172.26.76.194; 172.26.76.195	Private upstream routing
7	192.168.53.172	Private upstream routing
8	***	No response to traceroute probes
9	***	No response to traceroute probes

10	173.194.120.14; 72.14.217.206	Public upstream/internet hop
11	192.178.111.151; 192.178.110.227	Public upstream network
12	8.8.8.8 (dns.google)	Final destination

```

> traceroute 8.8.8.8
traceroute to 8.8.8.8 (8.8.8.8), 64 hops max, 40 byte packets
 1  reliance.reliance (192.168.29.1)  5.322 ms  4.829 ms  12.178 ms
 2  10.38.152.1 (10.38.152.1)  12.709 ms  10.078 ms  9.359 ms
 3  172.31.0.238 (172.31.0.238)  12.694 ms
    172.31.2.26 (172.31.2.26)  19.951 ms  16.543 ms
 4  172.28.75.32 (172.28.75.32)  9.867 ms
    192.168.53.188 (192.168.53.188)  9.083 ms
    172.28.75.26 (172.28.75.26)  29.521 ms
 5  172.26.76.214 (172.26.76.214)  18.300 ms  8.990 ms  7.727 ms
 6  172.26.76.194 (172.26.76.194)  17.122 ms  14.017 ms
    172.26.76.195 (172.26.76.195)  10.677 ms
 7  192.168.53.172 (192.168.53.172)  11.538 ms
    192.168.53.176 (192.168.53.176)  14.587 ms
    192.168.53.172 (192.168.53.172)  13.305 ms
 8  * * *
 9  * * *
10  173.194.120.14 (173.194.120.14)  15.866 ms *
    72.14.217.206 (72.14.217.206)  14.104 ms
11  192.178.111.151 (192.178.111.151)  13.555 ms
    192.178.110.227 (192.178.110.227)  13.849 ms *
12  dns.google (8.8.8.8)  16.757 ms  9.521 ms  13.741 ms
> █

```

Figure 14 — Complete traceroute output to 8.8.8.8.

The traceroute reached 8.8.8.8 after 12 hops. Two intermediate hops did not respond to the traceroute probes, but this did not stop the trace from reaching the final destination.

7. Switching Fundamentals — Domains & MAC Learning

Each active switch port acts as its own collision domain, while each VLAN creates a separate broadcast domain. In this design there are eight logical VLANs, so the intended setup has eight broadcast domains.

7.1 MAC Learning

When a frame arrives, the switch records the source MAC address and the port where it was received. If the destination MAC is already known, the frame is sent only to the required port; if it is unknown, the frame is flooded within that VLAN. These learned entries are eventually removed when they age out.

7.2 Practical Switching Verification

```
show mac address-table  
show vlan brief  
show interfaces status
```

The Packet Tracer records verified VLAN membership, active switch ports and Layer-2 forwarding. The redundant topology additionally demonstrated STP-controlled forwarding and blocking.

8. Switching Methods, Frame Format & MTU

8.1 Ethernet Frame Format

Field	Size	Purpose
Preamble + SFD	8 bytes	Synchronization and start-of-frame delimiter
Destination MAC	6 bytes	Layer-2 destination
Source MAC	6 bytes	Layer-2 source
802.1Q tag (optional)	4 bytes	VLAN identification on trunks
EtherType / Length	2 bytes	Identifies upper-layer protocol or payload length
Data / Payload	46–1500 bytes	Encapsulated network-layer data
FCS	4 bytes	Frame integrity check

8.2 Store-and-forward Latency

```
1500 bytes * 8 = 12,000 bits
12,000 / 1,000,000,000 = 0.000012 s = 12 μs
```

TechNova uses store-and-forward switching. In this method, the switch receives the complete frame and checks its FCS before sending it onward. This adds a very small delay, but the extra checking is a reasonable trade-off because reliable delivery is more important here than saving a few microseconds.

8.3 IPv4 Fragmentation

```
4000-byte IP packet, 1500-byte MTU, 20-byte IPv4
header Payloads: 1480 + 1480 + 1040 bytes
Fragments: 1500 + 1500 + 1060 bytes
Total: 3 fragments
```

8.4 Captured Ethernet Frame

The actual Wireshark Frame 36 had a length of 193 bytes (1544 bits), EtherType IPv4, and a TCP/TLS payload.

9. VLANs (Virtual LANs)

VLANs logically divide a switched network into separate broadcast domains. TechNova uses departmental and special-purpose VLANs to isolate sensitive traffic, control broadcasts and apply consistent policy.

9.1 Design VLAN Plan

VLAN	Purpose	Design gateway
10	HR / Admin	10.10.10.1
20	Finance	10.10.20.1
30	Sales / Marketing	10.10.30.1
40	Engineering / QA	10.10.40.1
50	Servers / IT	10.10.50.1
60	VoIP	10.10.60.1
70	Guest Wi-Fi	10.10.70.1
99	Management	10.10.99.1

9.2 Verified Packet Tracer VLAN Implementation

VLAN	Lab network	Gateway
10	192.168.10.0/24	192.168.10.1
20	192.168.20.0/24	192.168.20.1
30	192.168.30.0/24	192.168.30.1
40	192.168.40.0/24	192.168.40.1
999	192.168.99.0/24	192.168.99.1

The implemented lab assigned two PCs per departmental VLAN: Fa0/1–2 to VLAN 10, Fa0/3–4 to VLAN 20, Fa0/5–6 to VLAN 30 and Fa0/7–8 to VLAN 40.

```
vlan
  10
  name
    HR
vlan 20
  name
    FINANCE
vlan 30
  name
    SALES-MKT
vlan 40
  name ENGG-QA

interface range fa0/1-2
  switchport mode access
  switchport access vlan
  10

interface range fa0/3-4
  switchport mode access
  switchport access vlan
  20

interface range fa0/5-6
  switchport mode access
  switchport access vlan
  30

interface range fa0/7-8
  switchport mode access
  switchport access vlan
  40
```

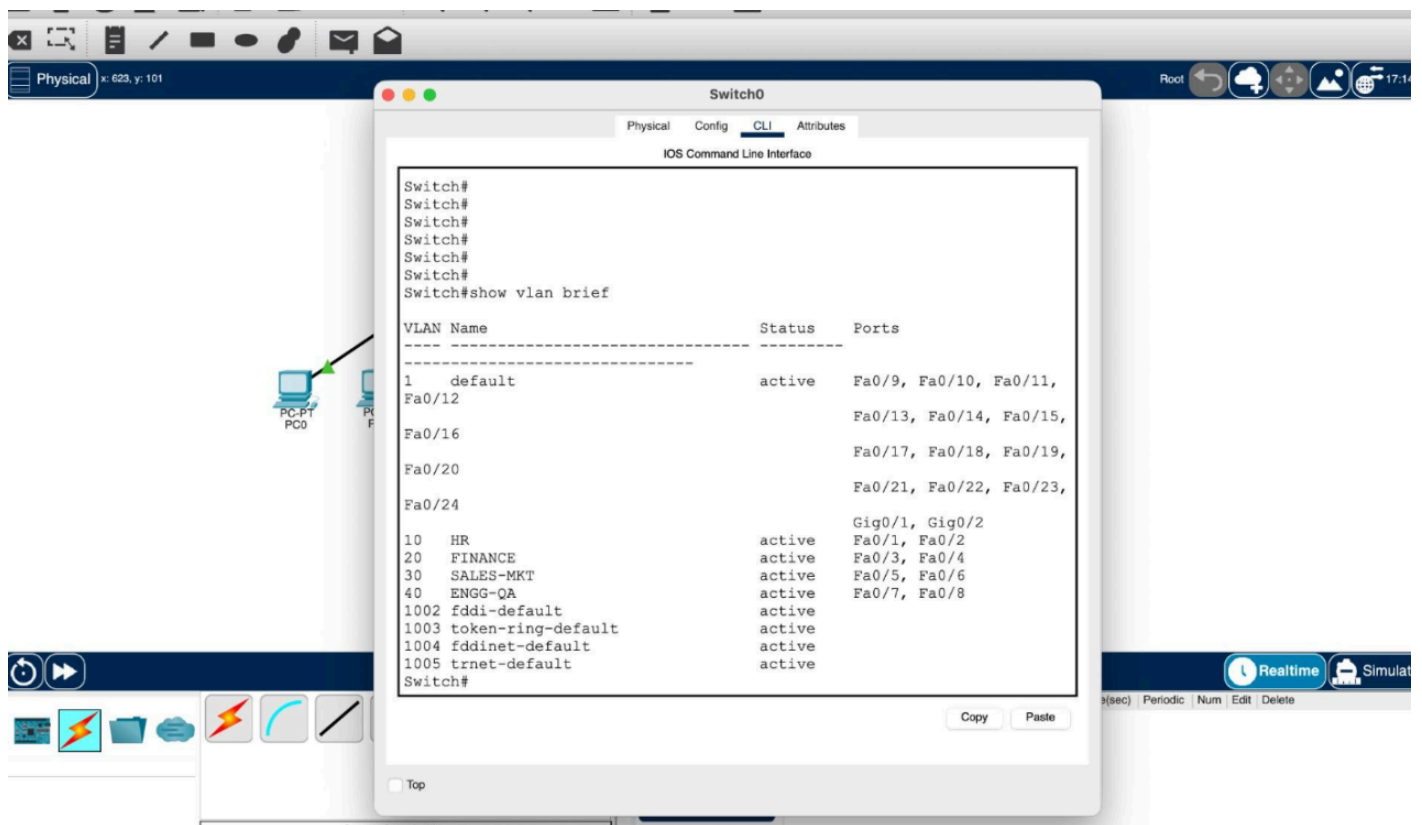


Figure 15 — VLAN configuration verification (show vlan brief).

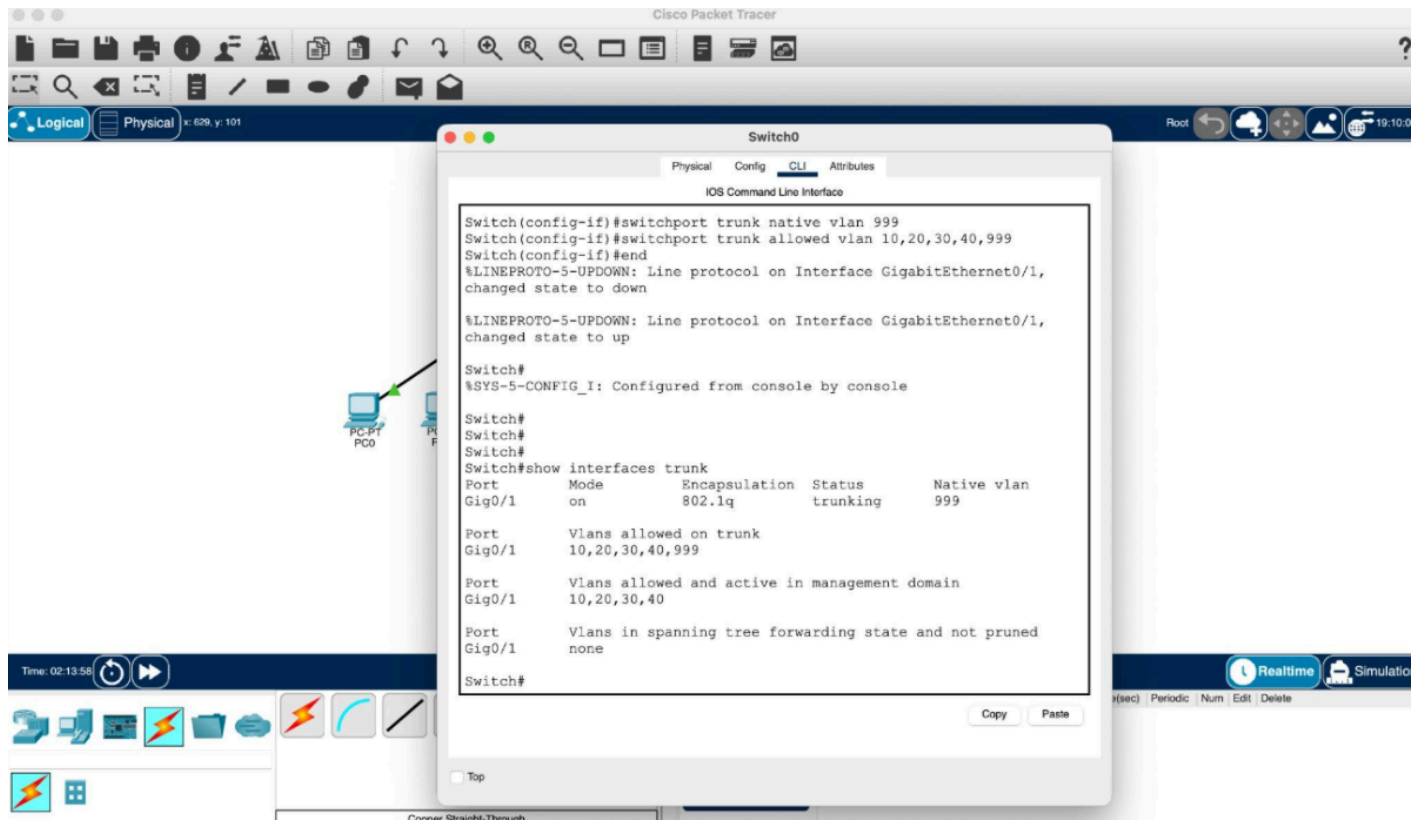


Figure 16 — 802.1Q trunk verification with native VLAN 999.

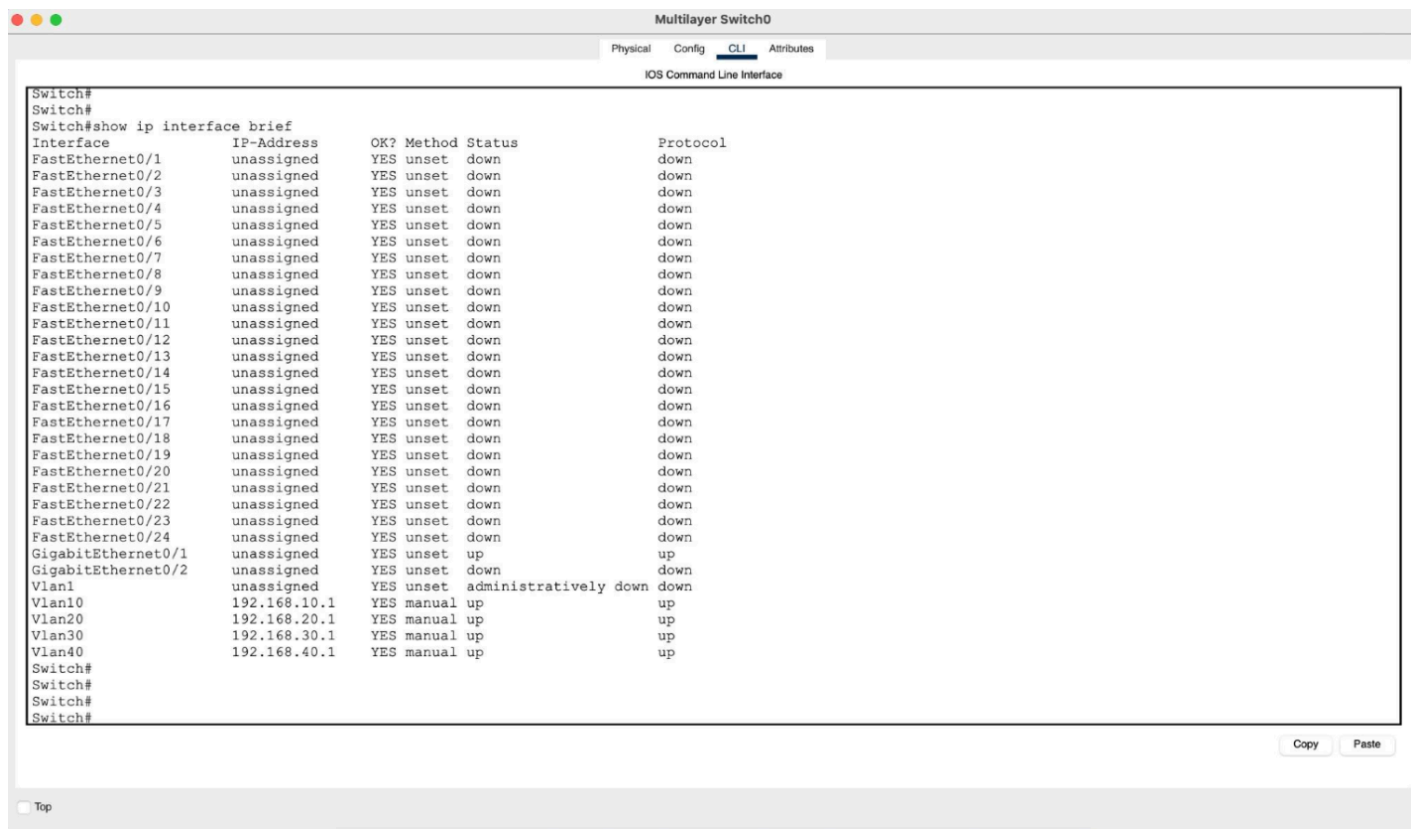


Figure 17 — SVI status showing VLAN 10/20/30/40 up/up.

10.802.1Q Trunking, Native VLAN & Layer-3 Switching

10.1 802.1Q Trunking

The Packet Tracer uplinks were configured as IEEE 802.1Q trunks. They carry VLANs 10, 20, 30 and 40, along with VLAN 999, which is used as the native VLAN.

```
interface gigabitEthernet 0/1
  switchport mode trunk
  switchport trunk native vlan 999
  switchport trunk allowed vlan 10,20,30,40,999
```

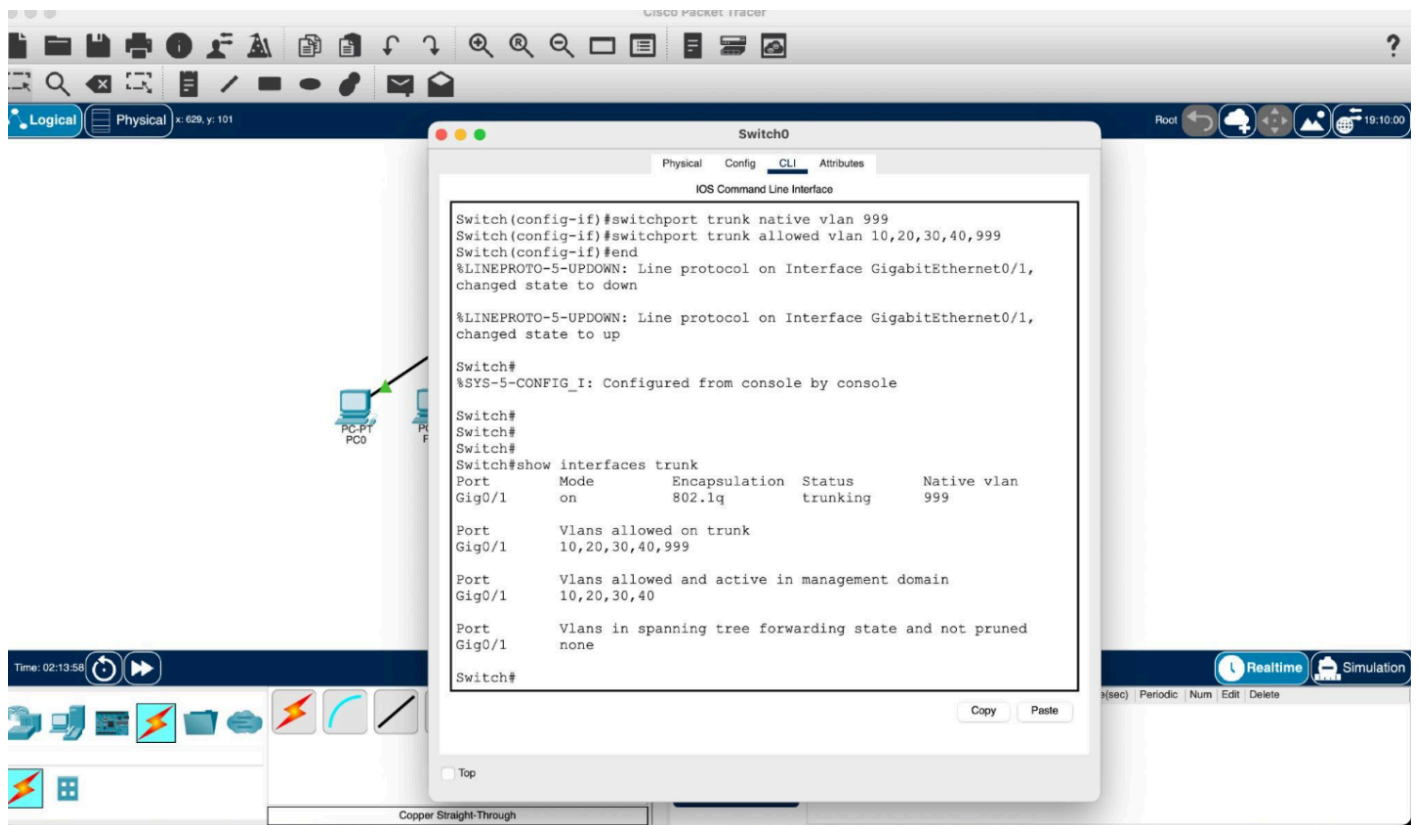


Figure 18 — Verified 802.1Q trunk with native VLAN 999 and allowed VLANs.

10.2 Native VLAN Mismatch and Correction

During the lab, a native VLAN mismatch was found and corrected by using VLAN 999 consistently on both sides of the trunk. Keeping the native VLAN the same avoids confusion when untagged frames cross the link.

```
interface vlan 10
ip address 192.168.10.1 255.255.255.0
no shutdown
interface vlan 20
ip address 192.168.20.1 255.255.255.0
no shutdown
interface vlan 30
ip address 192.168.30.1 255.255.255.0
no shutdown
interface vlan 40
ip address 192.168.40.1 255.255.255.0
no shutdown
```

```
ip routing
```

10.3 Layer-3 SVIs and Inter-VLAN Routing

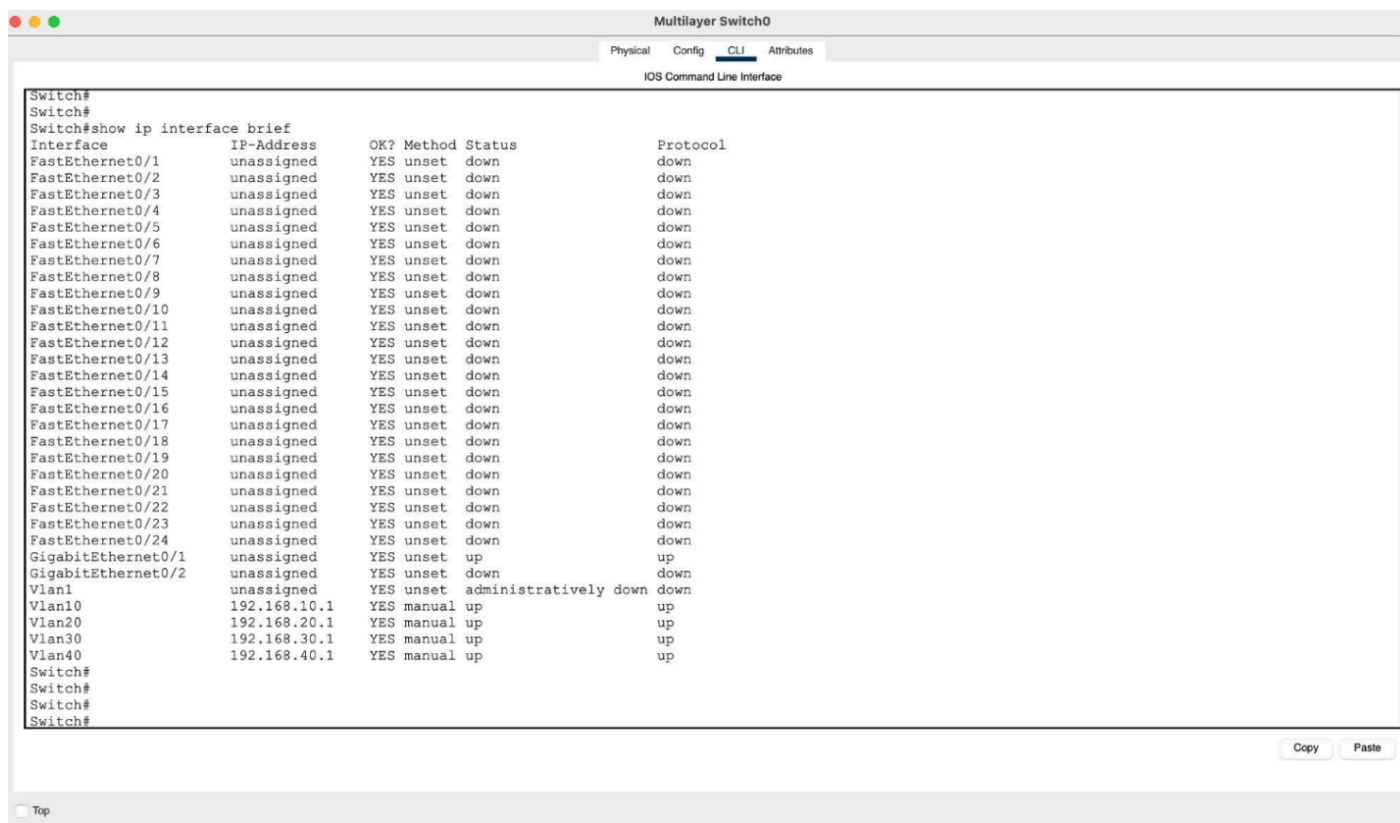


Figure 19 — SVI verification showing VLAN 10/20/30/40 up/up.

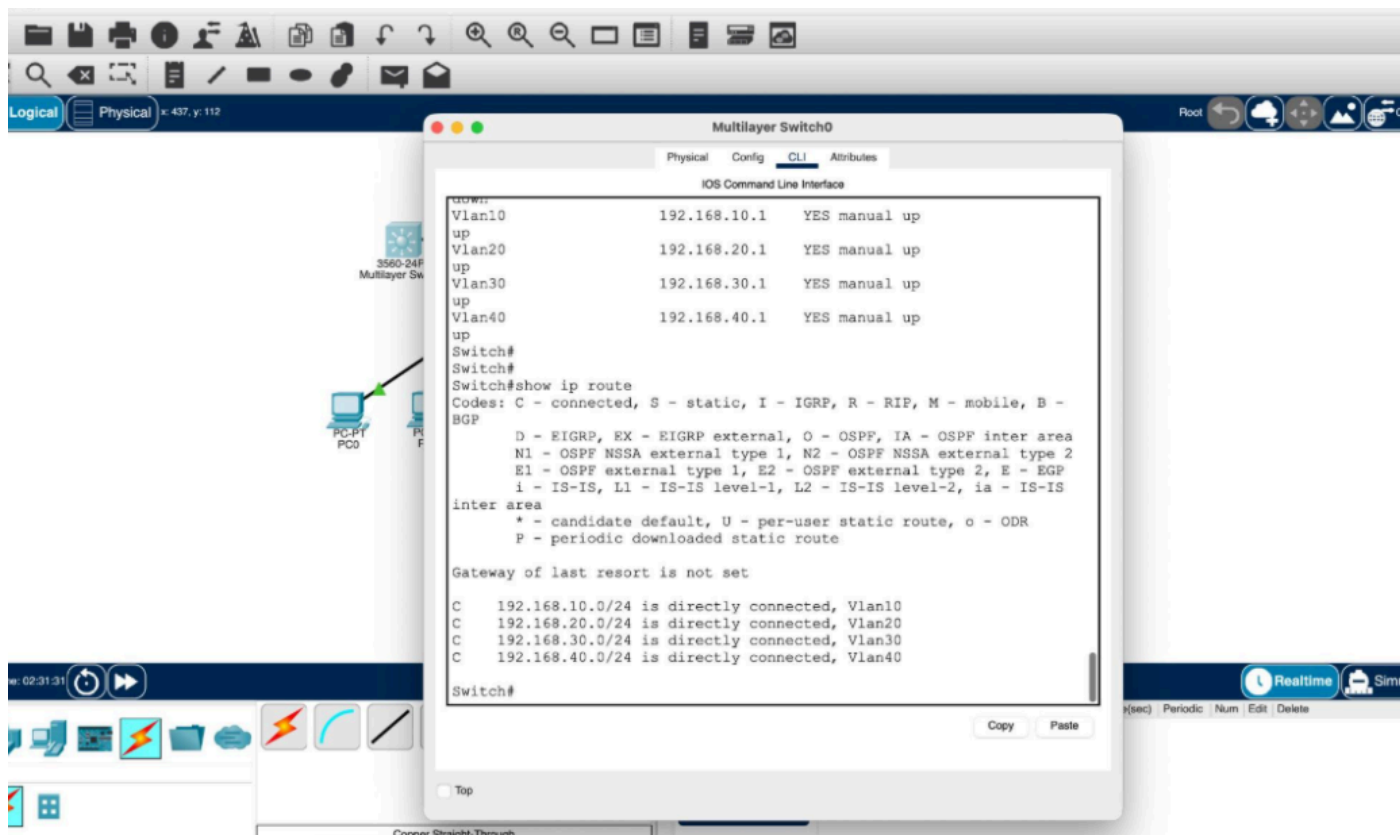


Figure 20 — Routing table showing the four directly connected VLAN networks.

The final routing table contained 192.168.10.0/24, 192.168.20.0/24, 192.168.30.0/24 and 192.168.40.0/24 as directly connected networks.

10.4 Inter-VLAN Connectivity Verification

```
ping 192.168.20.10
ping 192.168.30.1
ping 192.168.40.1
```

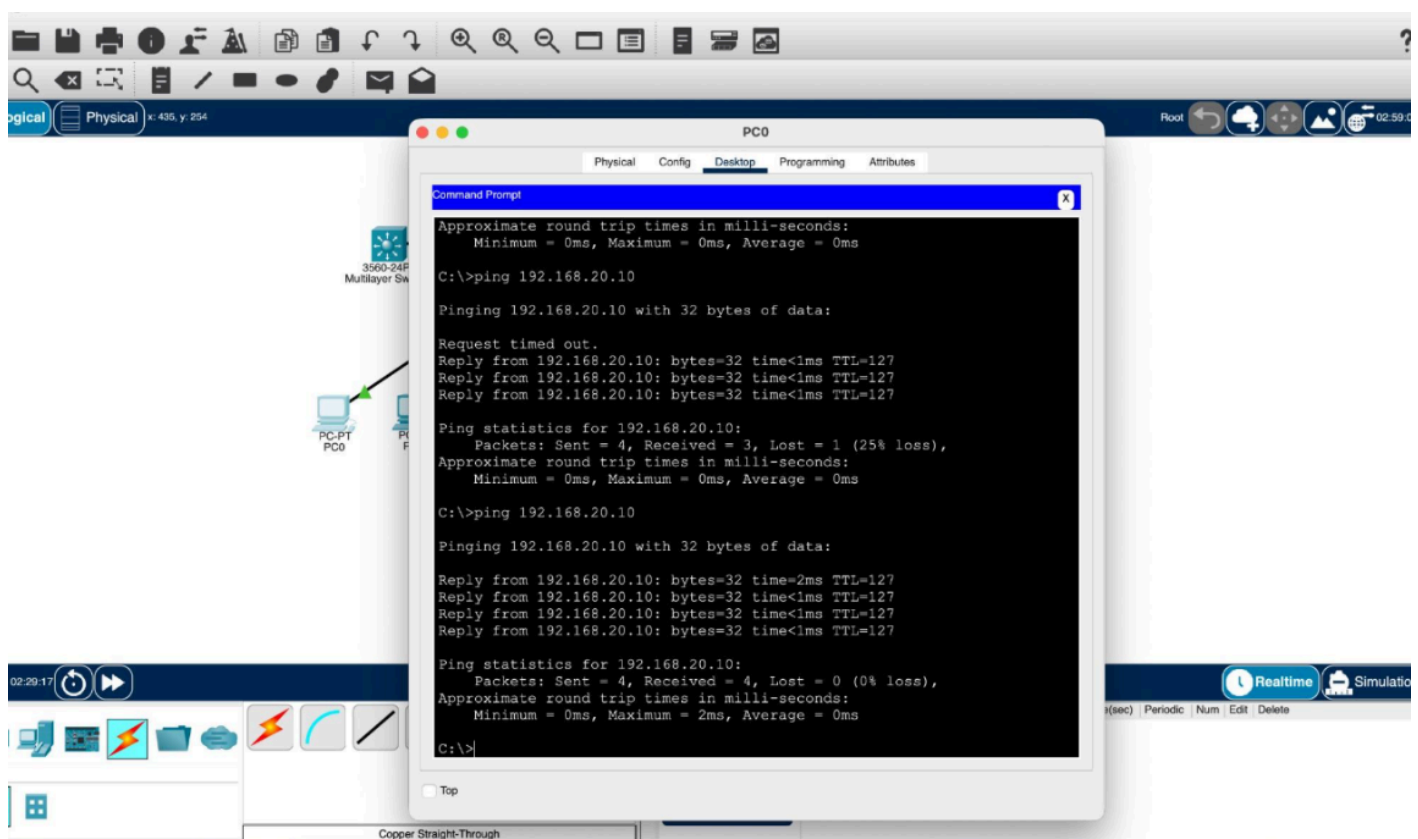


Figure 21 — Successful inter-VLAN ping to 192.168.20.10.

```

C:\>ping 192.168.30.1

Pinging 192.168.30.1 with 32 bytes of data:

Reply from 192.168.30.1: bytes=32 time<1ms TTL=255
Reply from 192.168.30.1: bytes=32 time<1ms TTL=255
Reply from 192.168.30.1: bytes=32 time<1ms TTL=255
Reply from 192.168.30.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.30.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.40.1

Pinging 192.168.40.1 with 32 bytes of data:

Reply from 192.168.40.1: bytes=32 time<1ms TTL=255
Reply from 192.168.40.1: bytes=32 time<1ms TTL=255
Reply from 192.168.40.1: bytes=32 time<1ms TTL=255
Reply from 192.168.40.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.40.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

```

Figure 22 — Successful connectivity tests to VLAN 30 and VLAN 40 gateways.

The final connectivity checks were successful. All four test pings received replies and showed 0% packet loss.

10.5 SVI-based Routing vs Router-on-a-stick

With router-on-a-stick, inter-VLAN traffic travels through router subinterfaces over one trunk, which can become a bottleneck as traffic increases. SVI-based Layer-3 switching performs the routing directly on the multilayer switch, making it a better fit for a larger internal network like TechNova.

10.6 Edge Routing and NAT

Connection	Address
Core SW Gi0/1	10.10.254.1/24
Edge Router G0/0/0	10.10.254.2/24

The edge router routes the four implemented internal networks via 10.10.254.1. NAT overload was configured with the core-facing interface as NAT inside and the WAN-facing

interface as NAT outside.

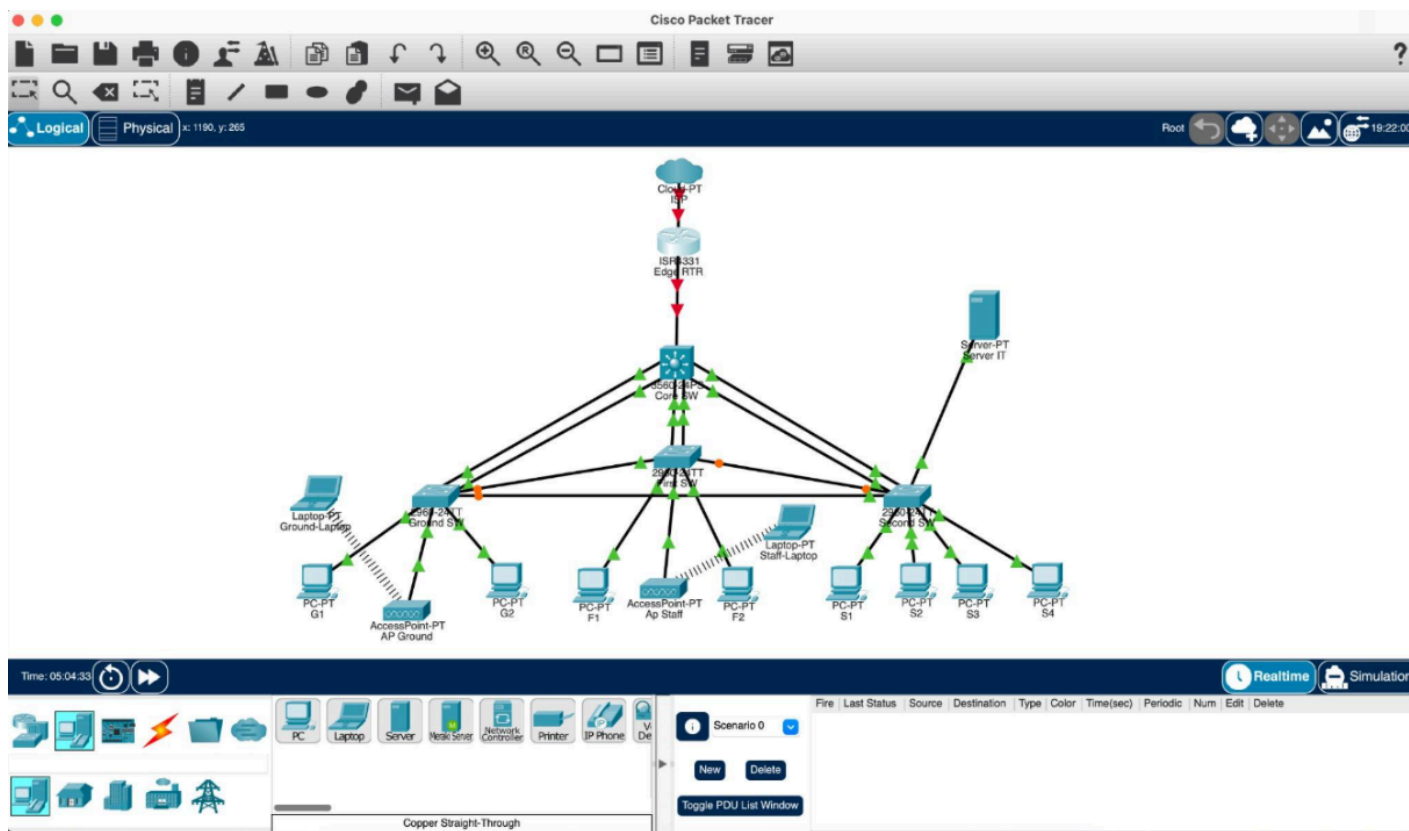


Figure 23 — Edge Router interface configuration.

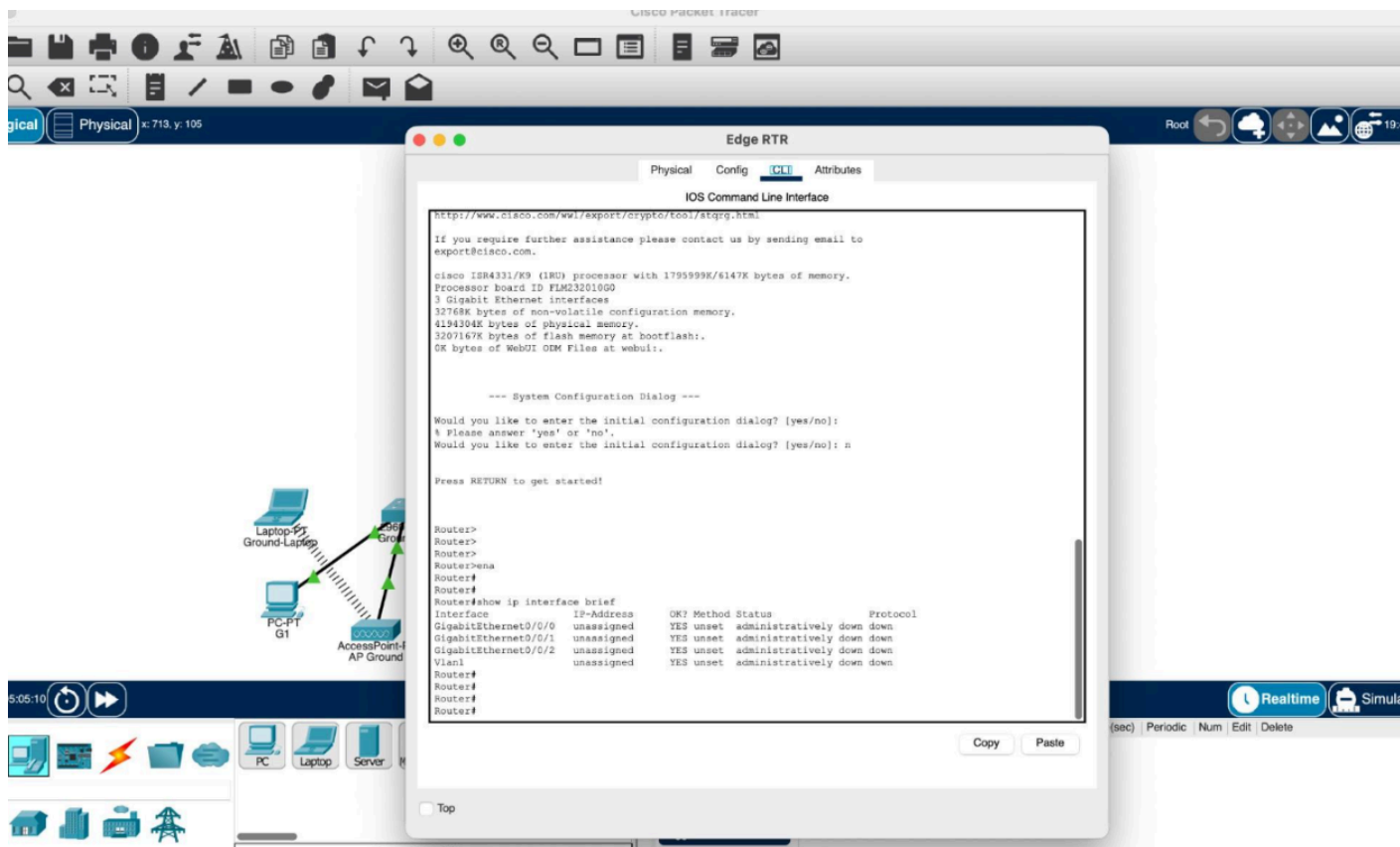


Figure 24 — Edge Router interface verification.

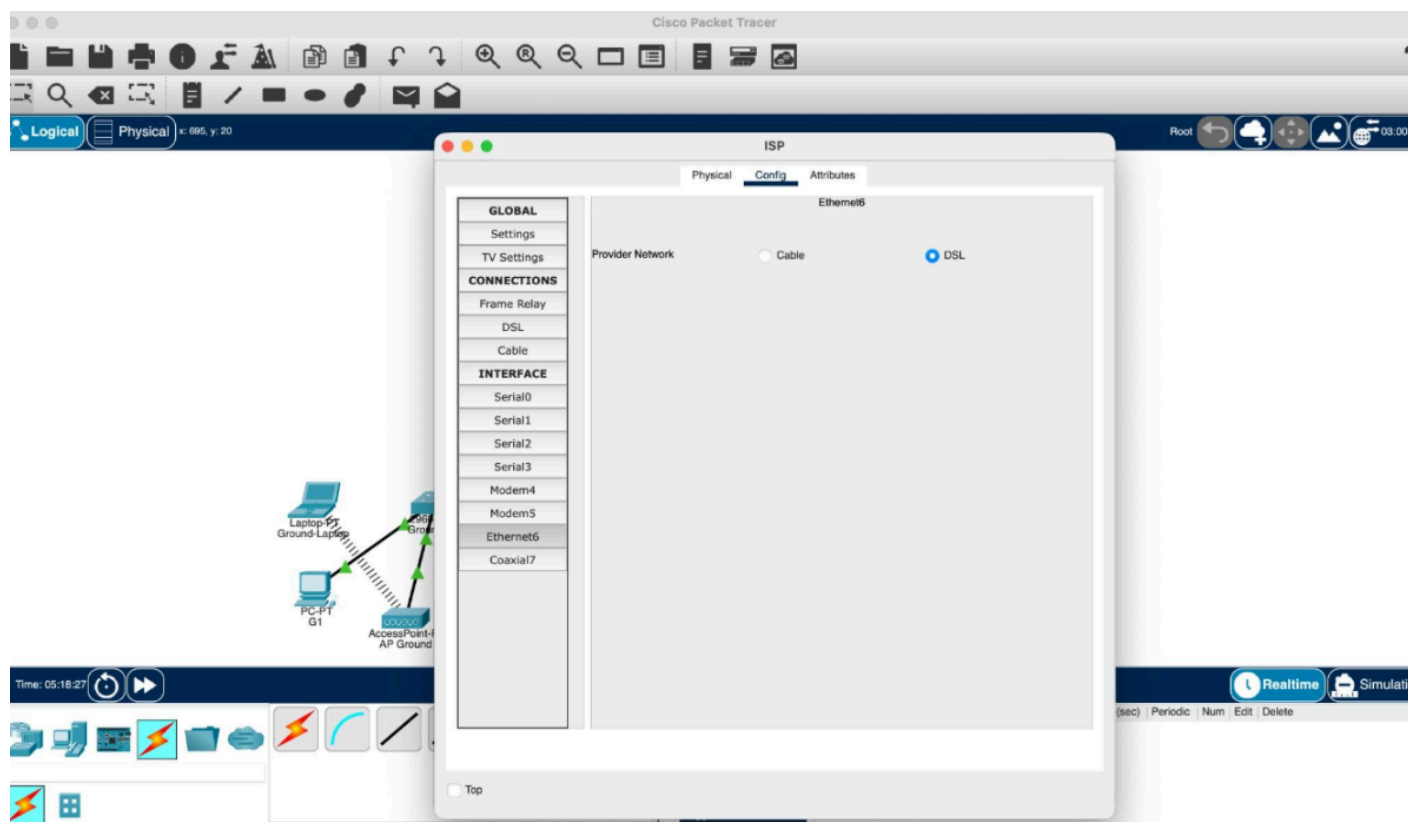


Figure 25 — NAT overload configuration.

11.Redundancy & Loop Prevention: STP / RSTP / PVST+

When multiple Layer-2 links are connected for backup, they can also create loops. RSTP handles this by keeping the extra path in a non-forwarding state until it is needed, which helps prevent broadcast storms while still keeping a backup connection available.

11.1 Root Bridge Selection

```
spanning-tree vlan 10,20,30,40,999 root primary
```

VLAN	Verified root status	Priority
10	This bridge is the root	24586
20	This bridge is the root	24596
30	This bridge is the root	24606
40	This bridge is the root	24616
999	This bridge is the root	25575

The Root ID address after configuration was 00E0.A307.347C, and the spanning-tree output explicitly reported RSTP as the active protocol.

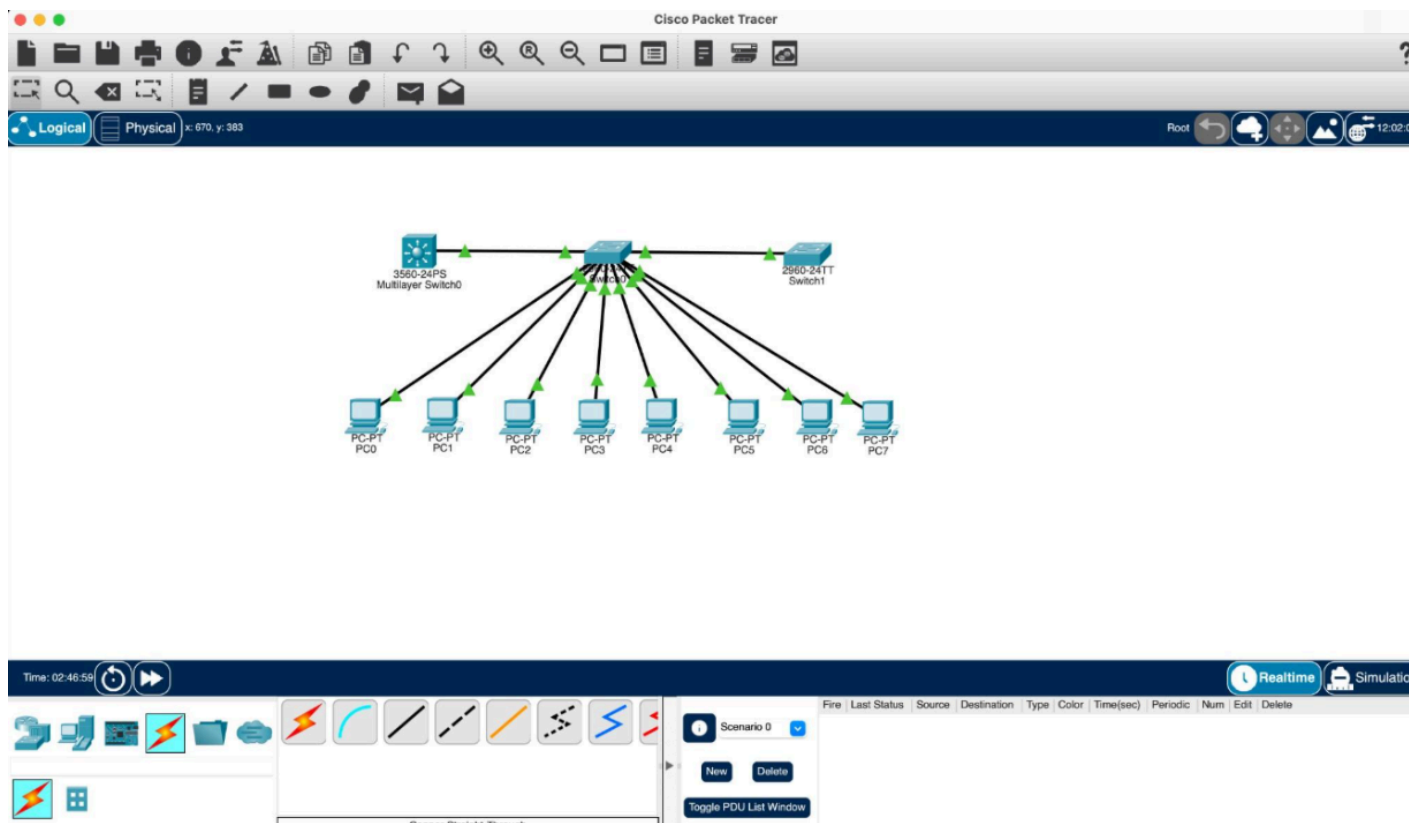


Figure 26 — Packet Tracer topology after routing and STP configuration.

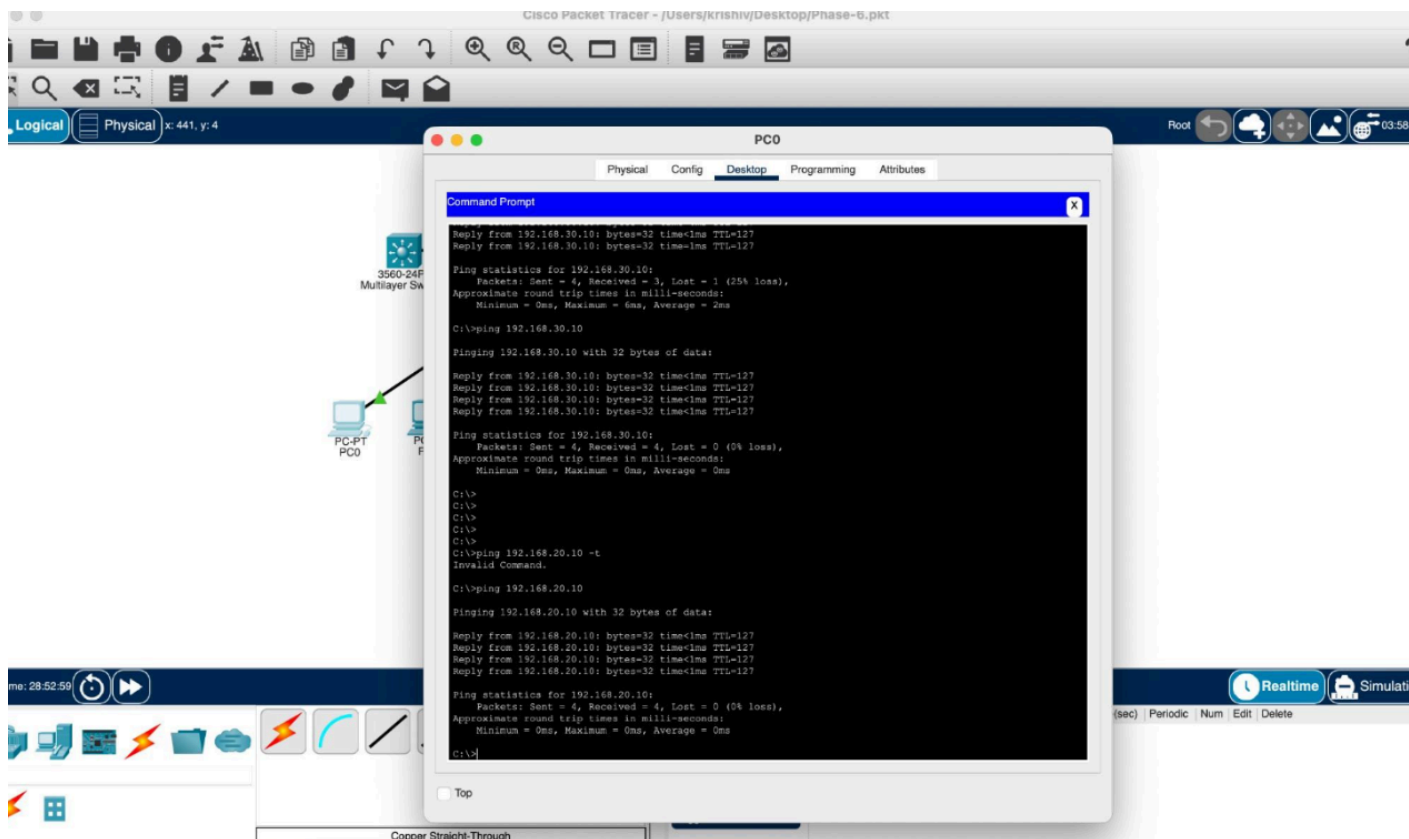


Figure 27 — Successful connectivity after STP configuration.

11.2 Redundant Topology and Loop Protection

The tested redundant setup creates a triangle between the multilayer switch, Switch0 and Switch1. The links between them are 802.1Q trunks carrying VLANs 10, 20, 30, 40 and 999.

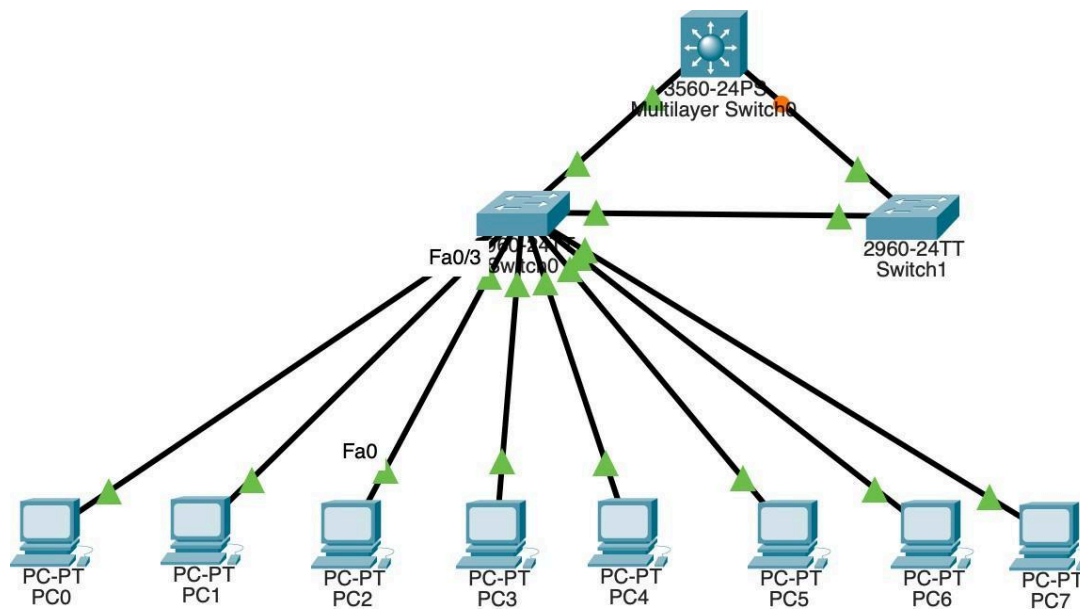


Figure 28 — Redundant-switch topology showing the triangular Layer-2 path.

11.3 RSTP Failure and Recovery

To check whether the redundancy actually worked, one active uplink on Switch1 was manually shut down. Before the failure, Gi0/2 was the main forwarding path and Gi0/1 was the alternate blocked path. After the shutdown, RSTP reconverged and moved traffic to Gi0/1. The path cost increased from 4 to 8, but connectivity was maintained with 0% packet loss.

```
interface gigabitEthernet 0/2
shutdown
```

Item	Before failure	After failure
Gi0/2	Root FWD	Administratively down
Gi0/1	Alternate BLK	Root FWD
Root path cost	4	8
Ping to 192.168.20.10	Successful	Successful
Packet loss	0%	0%



Figure 30 — VLAN-specific STP path-preference verification.

The VLAN-specific STP settings spread the preferred traffic paths across the two redundant uplinks. VLANs 10 and 20 prefer Gi0/1, while VLANs 30 and 40 prefer Gi0/2.

The earlier Phase 6 record did not include a separate physical-link failure test. The later Phase 8 work contains the actual controlled failure and recovery test, so that result is used as the evidence for failover.

12.Link Aggregation / EtherChannel & Load Balancing

EtherChannel bundles multiple physical links into one logical channel. It provides additional bandwidth and member-link redundancy while allowing STP to treat the bundle as one logical connection.

12.1 Verified EtherChannel

```
interface range gigabitEthernet 0/1-2
  channel-group 1 mode active
interface port-channel 1
  switchport mode trunk
  switchport trunk native vlan 999
  switchport trunk allowed vlan 10,20,30,40
```

In the final Packet Tracer configuration, Port-channels 1, 2 and 3 were verified as up/up. The trunks used 802.1Q, native VLAN 999 and allowed VLANs 10, 20, 30, 40 and 999.

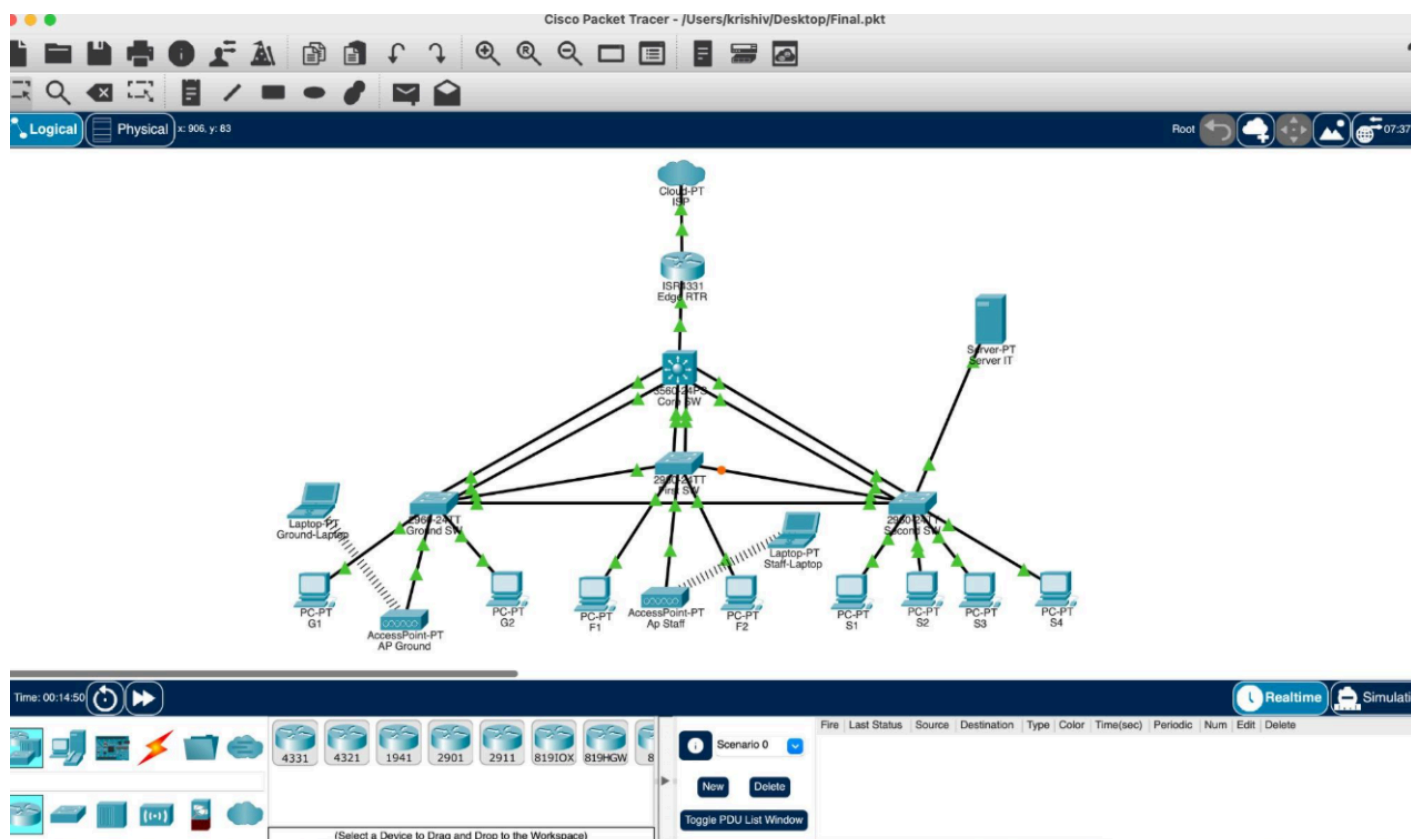


Figure 31 — Core Switch VLAN/routing verification from the final build.

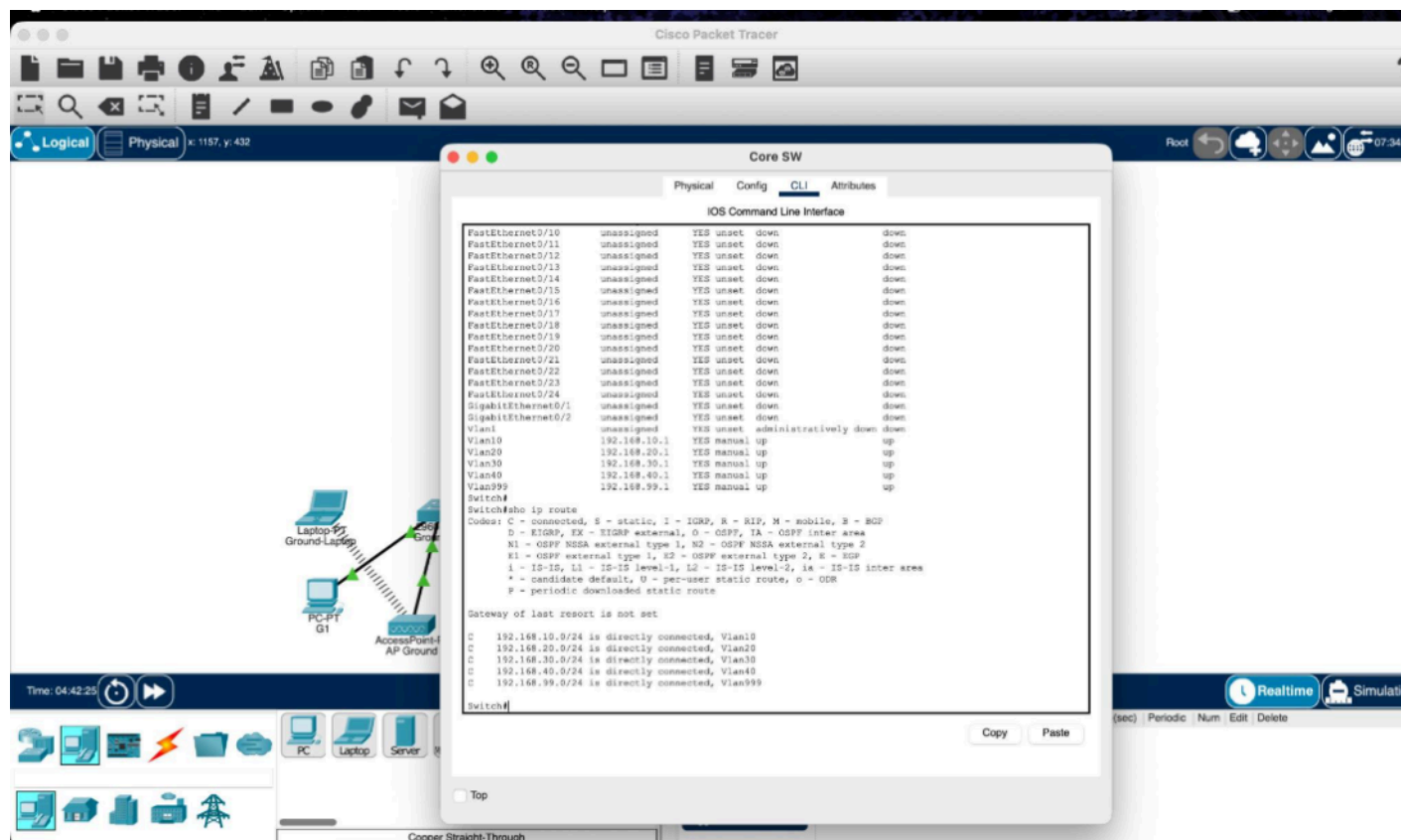


Figure 32 — CDP neighbor verification showing switch/router relationships.

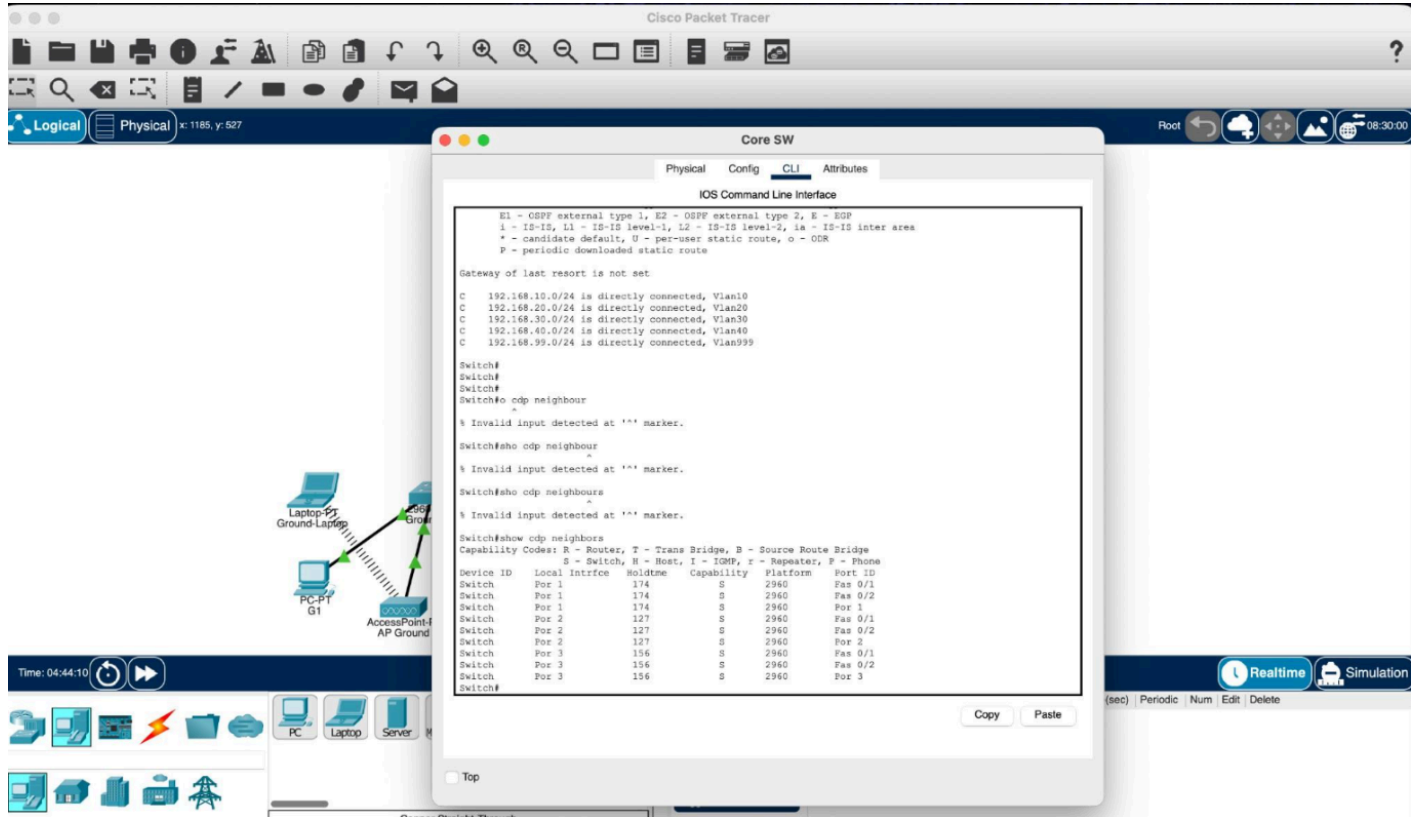


Figure 33 — Trunk / port status verification.

12.2 EtherChannel Failure and Mismatch Testing

The case-study task specifies member-failure and mismatch tests. The supplied final configuration verifies the EtherChannels as up/up, but it does not contain a recorded member-failure packet-loss result or a recorded mismatch/correction result. Those observations are therefore not claimed as completed evidence.

```

show etherchannel summary
show interfaces trunk
show spanning-tree
  
```

12.3 Final Connectivity Around the Aggregated Network

Test	Result
Server IT → 192.168.40.1	PASS — 0% loss
Server IT → 192.168.40.20	PASS
S3 → 192.168.40.20	PASS
S3 → 192.168.40.1	PASS
Core SW ↔ Edge Router	PASS

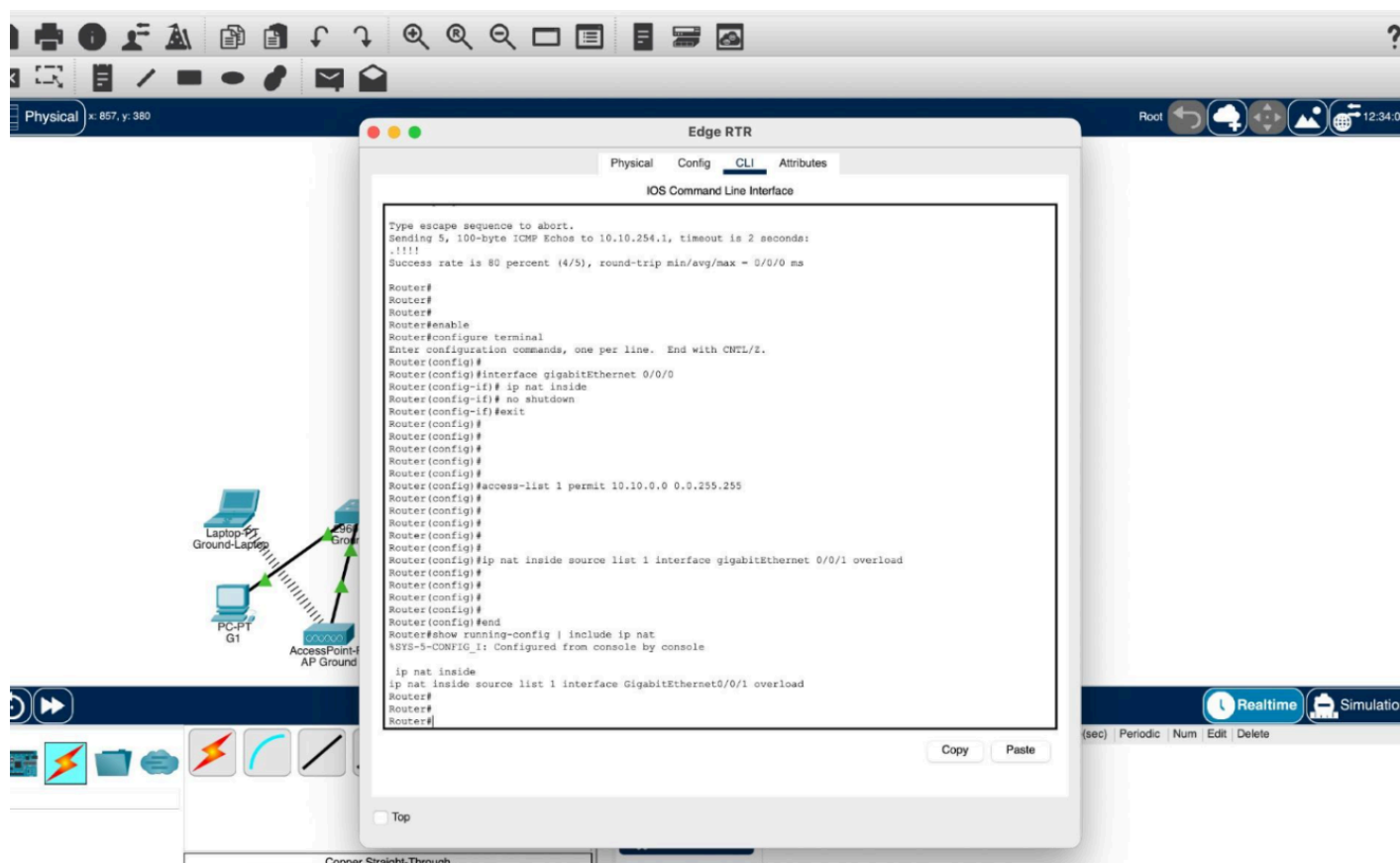


Figure 34 — Server connectivity test.

Command Prompt

```
Reply from 192.168.40.1: bytes=32 time<1ms TTL=255
Reply from 192.168.40.1: bytes=32 time<1ms TTL=255
Reply from 192.168.40.1: bytes=32 time<1ms TTL=255
Reply from 192.168.40.1: bytes=32 time=5ms TTL=255

Ping statistics for 192.168.40.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 5ms, Average = 1ms

C:\>ping 192.168.40.10

Pinging 192.168.40.10 with 32 bytes of data:

Reply from 192.168.40.10: bytes=32 time<1ms TTL=128
Reply from 192.168.40.10: bytes=32 time<1ms TTL=128
Reply from 192.168.40.10: bytes=32 time=14ms TTL=128
Reply from 192.168.40.10: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.40.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 14ms, Average = 3ms

C:\>ping 192.168.30.1

Pinging 192.168.30.1 with 32 bytes of data:

Reply from 192.168.30.1: bytes=32 time<1ms TTL=255
Reply from 192.168.30.1: bytes=32 time<1ms TTL=255
Reply from 192.168.30.1: bytes=32 time<1ms TTL=255
Reply from 192.168.30.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.30.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

Request timed out.
Request timed out.

Ping statistics for 10.0.254.1:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 10.0.254.1

Pinging 10.0.254.1 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 10.0.254.1:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```


Figure 35 — Final completed Packet Tracer topology.

13. Conclusion — The Design, End to End

Overall, the project brings together the TechNova network design and the practical Packet Tracer implementation. The final setup uses a hybrid topology, CIDR-based addressing, VLANs, 802.1Q trunks, Layer-3 SVIs, RSTP and EtherChannel to make the network organized, scalable and more reliable.

- The written design uses the 10.10.0.0/16 private address block with departmental subnets sized for the stated requirements.
- The practical Wireshark record verified Ethernet/IPv4/TCP/TLS encapsulation, ARP request/reply behavior, macOS IP configuration, ping performance and a 12-hop traceroute.
- The Packet Tracer implementation verified VLANs 10, 20, 30 and 40, native VLAN 999, 802.1Q trunking, SVI-based inter-VLAN routing and DHCP.
- The edge router provided the verified 10.10.254.1/24 ↔ 10.10.254.2/24 routed transit and NAT configuration for the implemented internal networks.
- RSTP was verified as active, the multilayer switch was made root for VLANs 10, 20, 30, 40 and 999, and a controlled link failure demonstrated successful failover with 0% packet loss.
- VLAN-specific STP preferences were verified so different VLANs use different preferred redundant uplinks.
- Port-channels 1, 2 and 3 were verified up/up in the final build.
- The ISP/Cloud portion was intentionally left outside the completed functional scope because the WAN interface did not obtain a DHCP lease; no arbitrary public address was assigned.
- The remaining production-level weakness is the single core Layer-3 switch; a second redundant core with first-hop/high-availability mechanisms would remove that single point of failure.

Final implementation status

The main network functions that were completed and verified include VLAN segmentation, native VLAN 999, inter-VLAN routing, EtherChannel, DHCP, core-to-edge routing, server connectivity, NAT configuration, RSTP root selection, the redundant topology and the controlled RSTP failover. Physical cable termination/testing, DNS A-record evidence, EtherChannel member-failure/mismatch testing and full ISP/Internet functionality were not completed or claimed.

Appendix A — Key Verification Commands

```
show ip interface brief
show ip route
show vlan brief
show interfaces trunk
show etherchannel summary
show spanning-tree
show ip dhcp binding
show cdp neighbors
```

Appendix B — Evidence Index

Figure	Evidence
1	Redundant-switch topology
2–5	Wireshark capture and Frame 36 analysis
6–7	ARP Request/Reply
8–10	DHCP configuration, client assignment and bindings
11-14	macOS IP configuration, DHCP information, ping and traceroute
15-17	VLAN, trunk and SVI verification
18-25	Trunking, SVI/routing, inter-VLAN tests, edge routing and NAT
26-30	STP/RSTP root selection, redundancy, failover and VLAN path preference
31-35	Final core verification, CDP, trunk status, server connectivity and final topology

All actual observations in this combined report are retained from the supplied practical records. Where a task was not performed or evidence was not supplied, the report states that explicitly rather than fabricating a result.

Final Topology

